



**MATHEMATICAL MODELLING AND ACTIVE VIBRATION CONTROL
OF A MOTOR GRADER OPERATOR CABIN USING PID AND LQR
METHODS**

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MECHATRONICS ENGINEERING MASTER'S THESIS**

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I hereby declare that all statements in this statement has been obtained and presented in accordance with academic rules and ethical conduct. I also declare that, as required by these rules and conduct, I have fully cited and referenced all material and results that are not original to this work.

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ABSTRACT

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TURACAN, MİHRİBAN

MECHATRONICS ENGINEERING MASTER'S THESIS

Supervisor: Asst. Prof. Dr. Ulaş BELDEK

Co-Supervisor: Dr. Hilal BİNGÖL

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Vibration was recorded on a Hidromek HMK 480MG grader at four points under three ground-engaging conditions. The campaign was repeated the same day with two support points removed, so that ground, operator and machine settings remained unchanged. Measured transmissibility falls from 0.62 to 0.57 through the mounting change alone.

A three-degree-of-freedom model was built from these measurements. Mount stiffness follows from the deflection of the load-bearing element under the cabin weight, and damping is the only quantity that had to be identified by fitting. Driving the model with the measured chassis spectrum reproduces the cabin response to within 4.5 per cent.

Two controllers were designed for an arrangement in which the removed mounts are replaced by actuators aligned with the centre of gravity, so that common-mode actuation yields pure vertical force. A PID-based law canceling ninety per cent of the transmitted mount force achieves a reduction of 57.5 per cent. A regulator built on absolute cabin acceleration achieves 91.4 per cent, the weighting proving decisive. Peak actuator demand is 0.60 kN, within commercial hydraulic capacity, and the overall transmissibility falls to 0.05.

Keywords: Active Vibration Control, Operator Cabin, Motor Grader, Linear Quadratic Regulator, Elastomeric Mount, Transmissibility.

ÖZET

PID VE LQR YÖNTEMLERİNİ KULLANARAK BİR MOTOR GREYDER OPERATÖR KABİNİNİN MATEMATİKSEL MODELLENMESİ VE AKTİF TİTREŞİM KONTROLÜ

TURACAN, MİHRİBAN
MEKATRONİK MÜHENDİSLİĞİ YÜKSEK LİSANS TEZİ

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Motor greyder operatörleri iş günü boyunca titreşime maruz kalmakta, kabini taşıyan elastomer takozlar ise bu titreşimi ancak kısmen sönümlemektedir. Bu çalışma, söz konusu titreşimin önce montaj düzenlemesi değiştirilerek, ardından aktif yolla ne ölçüde azaltılabileceğini araştırmaktadır.

Titreşim, Hidromek HMK 480MG greyder üzerinde dört noktada ve zemine temas eden üç çalışma koşulunda kaydedilmiştir. Ölçüm, iki destek noktasındaki takozlar sökülerek aynı gün tekrarlanmış, böylece zemin, operatör ve makine ayarları değişmeden kalmıştır. Yalnızca montaj değişikliğiyle ölçülen iletim oranı 0,62'den 0,57'ye düşmektedir.

Bu ölçümlerden üç serbestlik dereceli bir model kurulmuştur. Takoz sertliği, yük taşıyan elemanın kabin ağırlığı altındaki çökmesinden elde edilmiş; sönüm ise uydurma yoluyla belirlenmesi gereken tek büyüklük olmuştur. Model, ölçülen şasi spektrumuyla sürüldüğünde kabin tepkisini yüzde 4,5 hatayla yeniden üretmektedir.

Çıkarılan takozların yerine, ortak mod uyarımın saf dikey kuvvet üretmesi için ağırlık merkezi hizasına yerleştirilen aktüatörlerin kullanıldığı bir düzen için iki kontrolcü tasarlanmıştır. Aktarılan takoz kuvvetinin yüzde doksanını iptal eden PID tabanlı yasa yüzde 57,5 azaltma sağlamaktadır. Mutlak kabin ivmesi üzerine kurulan

reg lat r ise y zde 91,4 azaltma saėlamakta, aėırlıklandırmanın belirleyici olduėu g r lmektedir. En y ksek akt at r talebi 0,60 kN olup ticari hidrolik kapasitesi i indedir ve genel iletim oranı 0,05'e inmektedir.

Anahtar Kelimeler: Aktif Titreřim Kontrol , Operator Kabini, Motor Greyder, Doėrusal Karesel Reg lat r, Elastomer Takoz, İletim Oranı,

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TABLE OF CONTENTS

STATEMENT OF NONPLAGIARISM	III
ABSTRACT	IV
ÖZET.....	VI
ACKNOWLEDGEMENT	VIII
TABLE OF CONTENTS.....	IX
LIST OF TABLES	XIII
LIST OF SYMBOLS AND ABBREVIATIONS	XVI
CHAPTER I.....	1
INTRODUCTION.....	1
1.1 BACKGROUND INFORMATION	1
1.2 PROBLEM STATEMENT	2
1.3 AIM AND SCOPE	2
1.4 CONTRIBUTIONS AND PRACTICAL RELEVANCE	3
1.5 THESIS ORGANIZATION.....	4
CHAPTER II	6
LITERATURE REVIEW AND THEORETICAL BACKGROUND.....	6
2.1 WHOLE-BODY VIBRATION IN CONSTRUCTION MACHINERY	6
2.2 PASSIVE AND ACTIVE ISOLATION SYSTEMS	7
2.3 MULTI-DEGREE-OF-FREEDOM MODELLING OF CABINS	8
2.4 CONTROL STRATEGIES FOR VIBRATION SUPPRESSION	8
2.5 FUNDAMENTALS OF VIBRATION THEORY	9
2.5.1 The Single-Degree-of-Freedom Oscillator	9
2.5.2 Base Excitation and Transmissibility	9
2.5.3 Principles of Vibration Isolation	10
2.5.4 Human Response to Vibration.....	11
2.6 SYSTEMS WITH SEVERAL DEGREES OF FREEDOM.....	11
2.6.1 Equations of Motion	11
2.6.2 Natural Frequencies and Mode Shapes	12

2.6.3	State-Space Representation	12
2.7	ELASTOMERIC ISOLATION MOUNTS.....	13
2.8	CONTROL METHODS APPLIED IN THIS STUDY	13
2.8.1	Proportional and Derivative Action.....	13
2.8.2	The Linear Quadratic Regulator	14
2.9	NUMERICAL METHODS.....	15
CHAPTER III		16
SYSTEM MODELLING		16
3.1	THE MACHINE AND ITS CABIN MOUNTING	16
3.2	THE MOUNT ASSEMBLY AND ITS LOAD PATH.....	17
3.3	DETERMINATION OF MOUNT STIFFNESS	19
3.4	DETERMINATION OF THE DAMPING	19
3.5	EQUATION OF MOTION	21
3.5.1	Coordinates.....	21
3.5.2	Mount Deformation	22
3.5.3	Force and Moment Balance.....	22
3.5.4	Base Excitation.....	22
3.5.5	Expanded Form	23
3.6	SYSTEM MATRICES.....	23
3.7	SELECTION OF THE MOUNT CONFIGURATION	24
3.7.1	The Effect of Mount Count	24
3.7.2	Which Pair to Remove.....	25
3.8	MODAL PROPERTIES	25
3.9	STATIC DEFLECTION	26
3.10	MEASUREMENT CAMPAIGN	26
3.10.1	Instrumentation and Procedure.....	26
3.10.2	Data Reduction	27
3.10.3	Measurement Quality	27
3.10.4	Measured Results.....	28
3.11	EXCITATION FREQUENCY.....	28
3.12	OPERATING POINT	29
3.13	MODEL VALIDATION.....	30
3.13.1	Procedure	30

3.13.2	Results	30
3.13.3	The Six-Mount Prediction	31
CHAPTER IV.....		33
ACTIVE VIBRATION CONTROL DESIGN		33
4.1	THE PRINCIPLE OF ACTIVE ISOLATION	33
4.2	CONFIGURATIONS EXAMINED	34
4.3	ACTUATOR PLACEMENT	34
4.3.1	The Parasitic Pitch Moment	34
4.3.2	Alignment with the Centre of Gravity.....	35
4.3.3	Consequence for Pitch.....	35
4.4	PID-BASED CONTROL	36
4.4.1	The Control Law.....	36
4.4.2	Determination of the Integral Gain.....	36
4.4.3	Selection of the Cancellation Fraction	37
4.5	LINEAR QUADRATIC REGULATOR	38
4.5.1	Formulation of the Objective.....	38
4.5.2	Expression of the Output.....	39
4.5.3	Actuator Dynamics.....	39
4.5.4	Solution.....	40
4.6	SELECTION OF THE CONTROL WEIGHT	40
4.7	EVALUATION PROCEDURE	41
4.8	STABILITY	42
4.9	SELECTION OF THE CONTROL METHOD.....	42
CHAPTER V		44
RESULTS AND DISCUSSION		44
5.1	BASIS OF COMPARISON	44
5.2	CABIN ACCELERATION.....	44
5.3	TRANSMISSIBILITY	46
5.4	THE PASSIVE IMPROVEMENT.....	46
5.5	COMPARISON OF THE CONTROLLERS	47
5.6	CLOSED-LOOP BEHAVIOR.....	48
5.7	SENSITIVITY TO ACTUATOR BANDWIDTH.....	49
5.8	SUMMARY OF RESULTS.....	50

CHAPTER VI.....	51
CONCLUSIONS AND RECOMMENDATIONS.....	51
6.1 SUMMARY OF THE WORK	51
6.2 PRINCIPAL FINDINGS	51
6.3 RECOMMENDATIONS FOR FURTHER WORK	52
REFERENCES.....	54
APPENDICES	56
APPENDIX 1: MEASURED DATA.....	56
APPENDIX 2: COMPUTATIONAL IMPLEMENTATION.....	58

LIST OF TABLES

Table 1: Cabin Inertial Properties	16
Table 2: Mount Station Coordinates Relative to the Centre of Mass of the Cabin...	17
Table 3: Model Error for Damping Values Taken from the Literature.....	20
Table 4: Identification of the Damping Ratio	20
Table 5: Mount Properties.....	21
Table 6: Geometric Sums of the Mount Coordinates.....	23
Table 7: Lists the Principal Items for Both Configurations	24
Table 8: Effect of the Number of Mount Stations.....	24
Table 9: Consequence of Removing Each Rear Pair	25
Table 10: Modal Properties, Six-Mount Configuration	25
Table 11: Operating Conditions	27
Table 12: Measured Vertical Acceleration and Transmissibility.....	28
Table 13: Excitation Frequency by Condition	29
Table 14: Operating Point	29
Table 15: Model Validation, Eight-Mount Configuration	30
Table 16: Six-Mount Prediction.....	31
Table 17: Effect of Control Action, Six-Mount Configuration.....	33
Table 18: Effect of the Cancellation Fraction	37
Table 19: Effect of the Control Weight.....	40
Table 20: Alternative Control Methods	43
Table 21: Cabin Vertical RMS Acceleration [m/s^2].....	44
Table 22: Reduction from the Measured Baseline [%]	45
Table 23: Overall Transmissibility.....	46
Table 24: Measured Transmissibility by Configuration	46
Table 25: Controller Comparison.....	47
Table 26: Effect of Actuator Bandwidth, PID-Based Control	49
Table 27: Summary of Configurations.....	50
Table 28: Measurement Configuration	56

Table 29: Three-Axis Root-Mean-Square Acceleration, Eight Mounts [m/s ²].....	56
Table 30: Vertical Acceleration and Transmissibility	56
Table 31: Left–Right Balance Check	56
Table 32: Mount Deflection Measurement	57

LIST OF FIGURES

Figure 1: Transmissibility Against Frequency Ratio for Three Damping Ratios. The Threshold at $R = \sqrt{2}$ Is Independent of Damping.....	10
Figure 2: Mount Station Layout in Plan View and the Pair Marked R1 Lies Nearest the Centre of Gravity and Is the couple Taken out in the Six-Mount Configuration. 17	
Figure 3: Cross-Section of the Mount Assembly Showing the Load-Bearing Flange, the Lower Body and the Steel Sleeve	18
Figure 4: Mean Model Error Against the Damping Ratio; The Minimum Defines the Adopted Value	20
Figure 5: Measured Vertical Acceleration Spectra for the Eight-Mount Configuration. The Chassis Carries Substantive Data Above 100 Hz from the Engine, While the Cabin Response Is Concentrated Below 40 Hz	27
Figure 6: Measured Transmissibility by Condition and Configuration	28
Figure 7: Model Validation Against the Measured Cabin Acceleration, Eight-Mount Configuration. Percentages Give the Deviation.....	31
Figure 8: Actuator Coordinate. Positioning At the Vacated Mount Stations Produces a Pitch Moment; Alignment with the Centre of Gravity Gives Pure Vertical Force .	35
Figure 9: Reduction and Actuator Force Against the Cancellation Fraction.....	37
Figure 10: Reduction and Actuator Force Against the Control Weight, Swept Across Five Decades	41
Figure 11: Cabin Vertical Acceleration for Each Configuration and Condition	45
Figure 12: Reduction from the Measured Baseline.....	45
Figure 13: Closed-Loop Transmissibility and Control Moves the Effective Resonance from 7.83 Hz to 2.48 Hz.....	48
Figure 14: Time histories for the six-mount configuration under regulator control, coarse grading. The chassis acceleration is shown against the controlled cabin acceleration, and the actuator force below.	49
Figure 15: Effect of Actuator Bandwidth on the Reduction Achieved.....	50

LIST OF SYMBOLS AND ABBREVIATIONS

SYMBOLS

A	: State Matrix
B	: Control Input Matrix
B_u	: Actuator Distribution Matrix
c	: Damping Coefficient Per Mount Station
C	: Damping Matrix
C_z	: Output Matrix for Absolute Cabin Acceleration
d_i	: Deformation of Mount Station I
E_d	: Disturbance Input Vector
f_a	: Actuator Bandwidth
f_{exc}	: Excitation Frequency
f_n	: Natural Frequency
F_i	: Force Developed at Mount Station I
G	: State Feedback Gain Matrix
$H(f)$	: Transfer Function Magnitude
I_θ	: Pitch Moment of Inertia
I_φ	: Roll Moment of Inertia
J	: Cost Functional
k_{static}	: Static Stiffness Per Mount Station
k_{dyn}	: Dynamic Stiffness Per Mount Station
K	: Stiffness Matrix
m	: Cabin Mass
M	: Mass Matrix
n	: Number of Mount Stations
P	: Solution of the Algebraic Riccati Equation

q	: Vector of Generalized Coordinates
Q	: State Weighting Matrix
r	: Frequency Ratio
R	: Control Weighting Matrix
$S_{ch(f)}$	: Measured Chassis Acceleration Spectrum
T	: Transmissibility
u	: Control Input Vector
v_i	: Participation Vector of Mount Station I
W	: Static Load Per Mount Station
x	: State Vector
x_i, y_i	: Mount Station Coordinates
z	: Vertical Displacement Relative to the Chassis
z_b	: Absolute Vertical Displacement of the Chassis
Z	: Absolute Vertical Displacement of the Cabin
α	: Cancellation Fraction of the PID-Based Law
δ	: Static Deflection
ζ	: Damping Ratio
η	: Loss Factor
θ	: Pitch Rotation
ρ_u	: Control Weight of the Regulator
φ	: Roll Rotation
ω_a	: Actuator Bandwidth, Angular
ω_n	: Undamped Natural Angular Frequency

ABBREVIATIONS

CG	: Centre of Gravity
DOF	: Degree of Freedom
FFT	: Fast Fourier Transform
ISO	: International Organization for Standardization
LQR	: Linear Quadratic Regulator
PD	: Proportional Derivative
PID	: Proportional Integral Derivative
RMS	: Root Mean Square

ARE	: Algebraic Riccati Equation
WBV	: Whole-Body Vibration
SCADAS	: Signal Conditioning and Data Acquisition System
m	: Meter
N	: Newton
rad	: Radian
s	: Second
kg	: Kilogram
Hz	: Hertz
HMK	: Hidromek (Model Designation)

CHAPTER I

INTRODUCTION

1.1 BACKGROUND INFORMATION

Earthmoving machinery exposes its operators to mechanical vibration over long and uninterrupted working periods. Among the machines in this class the motor grader occupies a demanding position. Its function is to level ground with a blade held in direct contact with the soil, and the resistance encountered by that blade changes continuously as the machine advances. Each change spreads along the frame and reaches the operator from the cabin floor.

This type of exposure is not just a matter of comfort. Whole-body vibration transmitted through the seated posture has been associated with disorders of the lumbar spine. The standard ISO 2631-1[1] provides the chassis for quantifying such exposure: acceleration is passed through human vibration weighting filters that reflect the sensitivity of human body, which is greatest between approximately 4 and 12.5 Hz in the vertical direction.

The conventional means of vibration isolating an operator cabin is a set of mounts placed between the cabin and the frame. These components are cost-effective components, mechanically simple, require no external power and have high durability. Their behavior, however, is regulated by a single stiffness value, and that value must satisfy two requirements that pull in opposite directions. The mount must be stiff enough to carry the weight of the cabin without excessive static deflection, and soft enough to place the mounted resonance below the frequency band in which the machine operates. These requirements cannot both be met beyond a certain point.

Active vibration isolation offers a way past this limitation. If an actuator is introduced into the load path and commanded based on measured motion, the dynamic stiffness experienced by the cabin can be reduced without altering the static support that the passive mounts continue to provide. The two requirements are thereby separated, and each can be satisfied on its own terms.

1.2 PROBLEM STATEMENT

The machine examined in this study is a Hidromek HMK 480MG motor grader. Its cabin has a mass of 919 kg and is supported at eight stations, each containing an elastomeric mount assembly.

Vibration was measured at four points on the machine under three ground-engaging conditions. Averaged over these conditions, the chassis exhibits a vertical root-mean-square acceleration of 0.169 m/s^2 and the cabin 0.111 m/s^2 , giving a transmissibility of 0.62. The mounts therefore attenuate the vibration, but approximately two thirds of the chassis motion is still delivered to the operator.

An examination of the passive system explains why this figure cannot readily be improved. The vertical natural frequency of the cabin on eight mounts is 9.04 Hz, which places the onset of isolation at 12.79 Hz. The dominant excitation, obtained from the measured chassis spectrum, lies at 13.65 Hz, giving a frequency ratio of 1.51 against a threshold of 1.41. The system therefore operates within its isolation region, but only seven per cent above the threshold.

Two routes to a lower resonance exist and both encounter a limit. Removing mounts softens the system but raises the load carried by each remaining station towards the manufacturer's allowable value. Softening the mount itself has the same effect on the resonance but increases the static deflection, which varies inversely with the square of the natural frequency.

The problem addressed in this thesis is therefore the following: to what extent can the vertical vibration reaching the cabin be reduced, first by a change to the mounting arrangement and then by active means, and can such a reduction be achieved within actuator force and bandwidth limits that correspond to commercially available hardware?

1.3 AIM AND SCOPE

The aim of the study is to construct a dynamic model of the cabin and its mounting system from measurement, to validate that model against the measured response, and to design and compare controllers acting upon it.

The scope is defined by four decisions.

The cabin is represented as a rigid body with three degrees of freedom: vertical translation, pitch and roll. Structural flexibility of the cabin shell is not included, since the first elastic mode of a welded cabin structure of this size lies well above the

frequency range of interest and the mount stiffnesses are several orders of magnitude below the structural stiffnesses.

The control objective is the vertical cabin acceleration. Measurements on all three axes show the cabin response to be largest in the vertical direction, and the two actuators considered produce vertical force, so horizontal motion lies outside the reach of the proposed system in any case.

The mount stiffness is obtained from a measurement on the machine rather than from a catalogue value, and the excitation frequency is obtained directly from the measured chassis spectrum. The damping coefficient is the single parameter that could not be measured and was instead identified by fitting.

The study is theoretical and numerical. No prototype was constructed, and the control results represent what the model computes rather than what a physical installation has been observed to provide.

1.4 CONTRIBUTIONS AND PRACTICAL RELEVANCE

Active cabin isolation has been examined in the literature for agricultural vehicles, excavators and off-road machines, and the linear quadratic regulator method is a standard tool in that body of work. The contribution of the primary focus of this study lies in the treatment of several questions that arise when such a system is applied to a real-world machine.

The load path within the mounting system. Each station of the machine studied contains two resilient mounting elements, and the configuration in which they are installed determines which of them carries the cabin weight. By following the installation order, it was established that the vertical load passes through only the upper flange, the lower element sitting beneath the chassis bracket and completing the joint without contributing stiffness in the vertical direction. The stiffness was then obtained from the measured static deflection of that flange, which places the model on a measured basis rather than on a nominal rate.

A controlled contrast of two mounting configurations. Both configurations were measured on the same day with the same operator, the unchanged machine parameters and the same ground, so that the difference between them is traceable to the mounting change only. Transmissibility drops from 0.62 to 0.57 in every condition examined. This improvement requires no actuator, no power supply and no control, and the arrangement has since been taken up by the manufacturer's engineering teams.

The choice of which mounts to remove. Removing any symmetric pair reduces the vertical stiffness identically, so the vertical performance is the same in each case. The distinguishing quantity is the longitudinal balance of the remaining set. Removing the pair nearest the Centre of gravity reduces the residual imbalance from -242.8 mm to -10.2 mm and the heave–pitch coupling term by a factor of twenty-four, whereas removing either of the other pairs increases the imbalance.

The formulation of the regulator objective. In a base-excited isolation problem the state variables are relative displacements, and good isolation above resonance is characterized by a large relative displacement, since the cabin remains stationary while the chassis moves beneath it. A cost function that penalises relative displacement therefore drives the cabin to follow the chassis and destroys the isolation the mounts provide. The objective is instead constructed on absolute cabin acceleration. A related finding concerns the control weight: the same regulator structure delivers between 53 and 93 per cent reduction depending on the weight selected, so performance in such systems is governed as much by the care taken in weighting as by the choice of method.

Actuator placement. When both actuators occupy a longitudinal station displaced from the Centre of gravity, a common-mode command generates a parasitic pitch moment, and the pitch response grows rather than falling. Placing the actuators at the longitudinal station of the Centre of gravity reduces the sum of the moment arms to zero and eliminates the effect.

Several results bear on the feasibility of installation. The actuator represented a first-order lag rather than assumed ideal. The peak demand of 0.60 kN lies well within the capacity of commercial hydraulic linear actuators used on construction machinery. The six-mount arrangement raises the static load per station to 153.2 kgf, which is fifty-one per cent of the manufacturer's allowable value.

1.5 THESIS ORGANIZATION

Chapter II reviews the literature on operator vibration exposure, on the behavior of elastomeric isolators and on the control strategies applicable to base-excited systems and establishes the theoretical relations used in the remainder of the work.

Chapter III describes the measurement campaign, determines the model parameters from the measured data and manufacturer information, derives the

equations of motion and the modal properties, and validates the completed model against the measured cabin response.

Chapter IV presents the design of the two controllers, together with the configuration selection, the actuator placement, the actuator model and the determination of the tuning parameters.

Chapter V reports the results, comparing the configurations and the controllers across the three operating conditions and examining the actuator's demand and the closed-loop behavior.

Chapter VI states the conclusions, sets out the limitations of the study and identifies the work that should follow.

CHAPTER II

LITERATURE REVIEW AND THEORETICAL BACKGROUND

This chapter serves two purposes. The first sections survey the published work bearing on the problem: the exposure of construction machinery operators to whole-body vibration, the behavior of passive and active isolation systems, the modeling of cabins with several degrees of freedom, and the control strategies applicable to such systems. The remaining sections establish the theoretical relations used in the rest of the work.

2.1 WHOLE-BODY VIBRATION IN CONSTRUCTION MACHINERY

The exposure of equipment operators to whole-body vibration has been examined across a range of machine types, and the surveys consistently place earthmoving machinery at the severe end of the distribution. Kittusamy and Buchholz [2] reviewed the field measurements available for construction equipment and reported that motor graders produce vibration magnitudes among the highest recorded in that class, a result they attributed to the direct engagement of the blade with the ground and to the absence of intermediate compliance in the load path. Their review also drew attention to the combined effect of vibration and constrained posture, since the operator must maintain visual attention on the blade while seated for extended periods.

Blood, Rynell and Johnson [3] carried out field measurements on heavy equipment and reported frequency-weighted root-mean-square accelerations between 0.8 and 2.1 m/s² for operators of this class of machine, the range reflecting differences in task and in surface condition.

The relevance of these findings to the present work lies less in the absolute magnitudes, which depend on the weighting procedure and the assumed exposure duration, than in the identification of the frequency band that matters. The weighting curves defined by ISO 2631-1 [1] for vertical vibration emphasize the region between approximately 4 and 12.5 Hz, in which the seated human body exhibits its principal resonances, and attenuate contributions from higher frequencies. A cabin isolation

system that shifts energy out of this band therefore produces a benefit larger than the reduction in unweight acceleration alone would suggest.

2.2 PASSIVE AND ACTIVE ISOLATION SYSTEMS

Cabin mounts in construction machinery are conventionally rubber-to-metal assemblies whose stiffness and internal damping provide the isolation. Hostens and Ramon [4] characterized such mounts in the context of combine harvester cabins and noted that their effectiveness is confined to frequencies sufficiently above the mounted resonance, while at and below that resonance the mount amplifies rather than attenuates. This behavior is intrinsic to the passive element and cannot be removed by refinement of the material or the geometry.

This limitation arises because a single stiffness value must satisfy two simultaneous requirements. The mounting element supports the static weight of the cabinet, which sets a lower limit on stiffness to ensure deflection remains within acceptable levels. Additionally, it must position the mounted resonance below the excitation frequency band, which establishes an upper limit. For machines where the excitation frequency is close to the achievable resonance, these limits leave very little margin for tolerance.

Active systems eliminate the trade-off between these two requirements by incorporating an actuator into the load path. Elmadany and Abduljabbar [5] evaluated advanced suspension configurations for ride comfort in trucks, reporting that fully active systems improved isolation by up to 60% compared to passive systems, while semi-active systems fell somewhere in between. The improvement is obtained because the actuator responds to dynamic motion while the passive elements continue to carry the static load, so that the effective dynamic stiffness may be reduced without any change in static deflection.

Nguyen, Zhang, Le and Jiao [6] compared three cab isolation arrangements on an off-road vibratory roller and demonstrated that the choice of mount configuration alters both the magnitude and the distribution of the cabin response across its degrees of freedom. Their work is relevant here in that it establishes the configuration itself, and not only the control law applied to it, as a design variable.

2.3 MULTI-DEGREE-OF-FREEDOM MODELLING OF CABINS

A cabin supported at several points cannot be represented adequately by a single degree of freedom, because the mounts are distributed and the resulting motion couple's translation with rotation. Hostens and Ramon [4] developed a six-degree-of-freedom model of a tractor cabin suspension and validated it against measured frequency response functions, obtaining agreement sufficient to support design decisions.

Models of reduced order are widely used where the excitation and the control objective are confined to a subset of the possible motions. For grading operations, the dominant motions are vertical translation, pitch and roll, and three-degree-of-freedom representations have been applied successfully to such systems while retaining the computational economy required for control design [6]. The present study adopts this reduced representation, and Chapter III demonstrates that the roll degree of freedom decouples entirely from the remaining two under pure vertical base excitation.

2.4 CONTROL STRATEGIES FOR VIBRATION SUPPRESSION

Proportional–integral–derivative control remains the most widely applied structure in industrial practice, and Åström and Hägglund [7] set out both its design procedures and the circumstances under which each term contributes. Its attraction lies in the small number of parameters and in the physical interpretation attached to each. Its limitation, in the context of vibration isolation, is that the structure addresses a single input and a single output, so that a system with several actuators requires several loops whose interaction is not represented in the design.

The linear quadratic regulator provides systematic treatment of the multi-input case. Anderson and Moore [8] present the underlying theory, in which a quadratic cost functional in the state and the control is minimized subject to the linear dynamics, yielding a state feedback gain obtained from the solution of an algebraic Riccati equation. The method accommodates any number of inputs and outputs, incorporates the control effort explicitly through the weighting on the input, and guarantees closed-loop stability when the weighting matrices satisfy the customary definiteness conditions.

Comparative studies of the two approaches applied to vehicle suspensions have been reported. Nagarkar, Bhalerao, Vikhe Patil and Zaware Patil [9] examined a nonlinear quarter-car suspension under both PID and LQR control and reported that

the relative performance depends on the objectives assigned to each, a conclusion consistent with the finding reported in Chapter IV of the present work concerning the sensitivity of the regulator to its weighting.

Robust methods, and in particular H-infinity synthesis, have also been applied to cabin suspension problems and offer guaranteed performance in the presence of bounded model uncertainty. Their application requires a quantitative description of that uncertainty, which was not available for the mount assembly examined here, and they are accordingly identified in Chapter VI as a direction for subsequent work.

2.5 FUNDAMENTALS OF VIBRATION THEORY

2.5.1 The Single-Degree-of-Freedom Oscillator

For a mass m supported on a spring of stiffness k and a viscous damper of coefficient c , free motion is governed by

$$m \ddot{z} + c \dot{z} + k z = 0 \quad (2.1)$$

Rao [10] gives the standard solution. In the absence of damping the mass oscillates at the undamped natural frequency

$$\omega_n = \sqrt{\frac{k}{m}}, \quad f_n = \omega_n / 2\pi \quad (2.2)$$

Damping is expressed non-dimensionally by comparing the coefficient present in the system with the value that would just prevent oscillation,

$$\zeta = c / (2 \sqrt{k m}) \quad (2.3)$$

and the damped natural frequency follows as $\omega_d = \omega_n \sqrt{1 - \zeta^2}$.

2.5.2 Base Excitation and Transmissibility

The cabin problem is one of base excitation: the support moves and the response of the mounted mass is sought. Writing q for the displacement of the mass relative to the base, the equation of motion becomes

$$m \ddot{q} + c \dot{q} + k q = -m \ddot{z}_b \quad (2.4)$$

in which $\ddot{z}_b$ denotes the acceleration of the base. The term on the right is not a physical force but an inertial term arising from the transformation to the relative coordinate; its derivation is set out in Section 3.5.

For harmonic base motion the ratio of the absolute acceleration of the mass to that of the base defines the transmissibility. Introducing the frequency ratio $r = \frac{f}{f_n}$,

$$T(r) = \sqrt{\frac{[1 + (2\zeta r)^2]}{[(1 - r^2)^2 + (2\zeta r)^2]}} \quad (2.5)$$

Equation (2.5) is central to the present work. It is applied both to individual frequency lines, where it describes how much of the chassis motion at that frequency reaches the cabin, and in the aggregate, where the predicted cabin response follows from the measured chassis spectrum.

2.5.3 Principles of Vibration Isolation

Three regions may be distinguished in the transmissibility characteristic. For r well below unity the mass follows the base and T approaches one, no isolation being obtained. In the neighborhood of r equal to one the system is in resonance, and T may exceed unity by a large factor. For r above a certain threshold the transmissibility falls below unity and isolation is achieved.

The threshold follows from setting T equal to unity in Equation (2.5). The condition reduces to $(1 - r^2)^2 = 1$, whose positive root is

$$r = \sqrt{2} \quad (2.6)$$

The damping ratio is cancelled from this result, so that the threshold is independent of damping. This yields the fundamental design rule of passive isolation: a mount attenuates only when the excitation frequency exceeds $\sqrt{2}$ times the mounted natural frequency and amplifies below it.

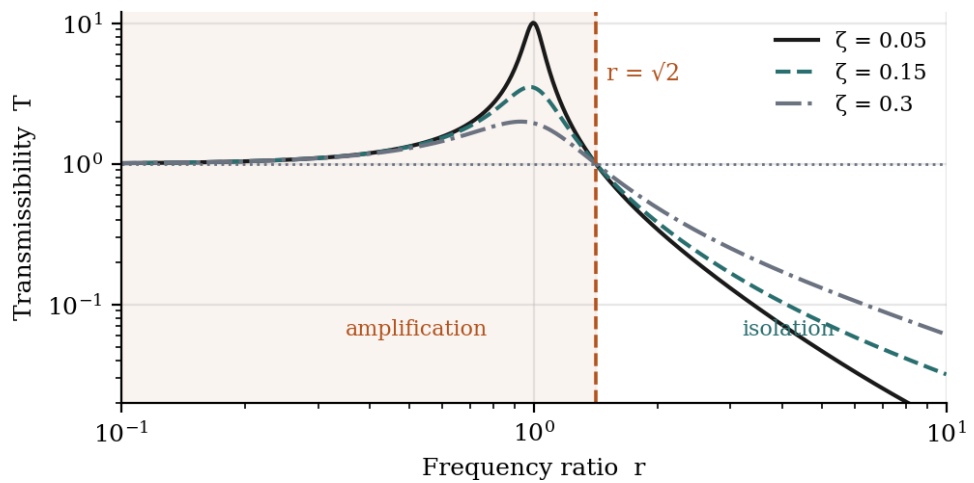


Figure 1: Transmissibility Against Frequency Ratio for Three Damping Ratios. The Threshold at $R = \sqrt{2}$ Is Independent of Damping

The rule imposes the constraint identified in Chapter I. Lowering f_n widens the isolation region, but the static deflection under self-weight is

$$\delta = m g / k \quad (2.7)$$

Combining Equations (2.2) and (2.7) gives δ proportional to $1/f_n^2$. A reduction of the natural frequency by a factor of two therefore increases the static deflection by a factor of four.

2.5.4 Human Response to Vibration

ISO 2631-1 [1] defines the procedure by which vibration exposure is evaluated for the seated human. Acceleration is recorded where the body meets its support and then filtered so that each frequency contributes in proportion to the sensitivity of the body at that frequency. Separate filters apply to the vertical direction, denoted W_k , and to the two horizontal directions, denoted W_d . The weighted result is normalized to an eight-hour reference period to give the daily exposure quantity, which is compared with the action value of 0.5 m/s² and the limit value of 1.15 m/s².

The present study reports unweighted root-mean-square acceleration throughout, for two reasons. The first is that the objective is the comparison of configurations and controllers against a common baseline, for which the unweighted quantity is a consistent measure; since the weighting is a linear operation applied identically to every case, the relative reductions reported in Chapter V would be unchanged by its application. The second is that the measured exposure lies well below the action value, so the objective of the work is improvement rather than compliance.

2.6 SYSTEMS WITH SEVERAL DEGREES OF FREEDOM

2.6.1 Equations of Motion

A rigid body supported at several points and possessing n degrees of freedom is governed by

$$M \ddot{q} + C \dot{q} + K q = f \quad (2.8)$$

The three matrices carry the inertial, dissipative and elastic properties of the body respectively, q collects the coordinates chosen to describe its configuration, and f collects the forces conjugate to those coordinates. Shabana [11] develops the systematic construction of these matrices for multibody systems.

2.6.2 Natural Frequencies and Mode Shapes

If both the dissipation and the external forcing are suppressed and the motion is taken to be harmonic, $q = V \sin(\omega t)$, then Equation (2.8) collapses to an eigenvalue problem in the squared frequency,

$$(K - \omega^2 M) V = 0 \quad (2.9)$$

A solution other than the trivial one requires the coefficient matrix to be singular, so that

$$\det(K - \omega^2 M) = 0 \quad (2.10)$$

and the roots of this determinant are the squared natural frequencies. Each root carries an associated eigenvector, which describes the shape the structure assumes when vibrating in that mode. Where the off-diagonal terms of K are small in comparison with the diagonal terms, the modes are weakly coupled, and each is dominated by a single coordinate.

The modal damping ratio of the i -th mode follows from the eigenvector and the damping matrix,

$$\zeta_i = (V_i^T M^{-1} C V_i) / (2 \omega_i V_i^T V_i) \quad (2.11)$$

2.6.3 State-Space Representation

Control synthesis requires dynamics in first-order form. This is achieved by treating the velocities as additional unknowns alongside the coordinates, which doubles the dimension but removes the second derivative,

$$x = [q; \dot{q}] \quad (2.12)$$

whose derivative is obtained by solving Equation (2.8) for the acceleration. The result is

$$\dot{x} = A x + B u + E_d w \quad (2.12)$$

With

$$A = \begin{bmatrix} 0_{3 \times 3} & I_{3 \times 3} \\ -M^{-1}K & -M^{-1}C \end{bmatrix}, \quad B = \begin{bmatrix} 0_{3 \times 2} \\ M^{-1}B_u \end{bmatrix} \quad (2.13)$$

in which u is the vector of control inputs, B_u distributes those inputs among the degrees of freedom, and E_d distributes the disturbance w .

The transfer function from disturbance to output follows by assuming harmonic excitation. For an output $y = C_z x$, the magnitude of the response at frequency f is

$$|H(f)| = |C_z(j2\pi f I - A)^{-1} E_d| \quad (2.15)$$

Equation (2.15) is the form in which all transmissibility values reported in this thesis are computed.

2.7 ELASTOMERIC ISOLATION MOUNTS

The Kelvin–Voigt representation is customary for rubber isolators. It places an elastic element and a dissipative element side by side across the same deformation, so that the total force carried by the mount is the sum of a term proportional to deformation and a term proportional to its rate,

$$F = k d + c \dot{d} \quad (2.16)$$

where d is the deformation. The model is an idealization of a material whose behavior is viscoelastic, and two consequences require attention.

The first is that the stiffness of rubber depends on the rate at which it is loaded. A quasi-static load–deflection measurement yields the static rate, whereas a component excited at vibration frequencies exhibits a higher dynamic rate. The ratio between the two is a material property, and values between 1.1 and 1.6 are reported for the elastomers used in mount applications. Vibration analysis requires dynamic value; the static value governs only the deflection under self-weight.

The second is that the energy dissipated by rubber per cycle is approximately independent of frequency, a behavior described as hysteretic and characterized by the loss factor η . An equivalent viscous coefficient may be defined at a chosen frequency through

$$\eta = c \frac{\omega}{k} \quad (2.17)$$

Loss factors between 0.05 and 0.15 are commonly quoted for the natural and synthetic rubbers used in cabin mounts. Section 3.4 shows that values in this range do not reproduce the response measured on the machine studied here and describes the procedure by which the damping was instead identified.

2.8 CONTROL METHODS APPLIED IN THIS STUDY

2.8.1 Proportional and Derivative Action

A control law that generates a force proportional to a measured displacement and its derivative,

$$u = K_p q + K_d \dot{q} \quad (2.18)$$

is described as proportional derivative. The proportional term acts as an additional stiffness and the derivative term as an additional damping, and the two gains may therefore be interpreted directly in mechanical terms. Åström and Hägglund [7] discuss the role of the integral term, which removes steady-state error where the plant does not itself provide a restoring action. Section 4.4 examines whether that condition obtains in the present application and determines the appropriate value of the integral gain accordingly.

2.8.2 The Linear Quadratic Regulator

The regulator returns the feedback gain for which the following quadratic measure of performance attains its smallest value,

$$J = \int_0^{\infty} (x^T Q x + u^T R u) dt \quad (2.19)$$

subject to the dynamics of Equation (2.13) with the disturbance absent. The matrix Q is positive semi-definite and weights the states, while R is positive definite and weights the control effort. Anderson and Moore [8] show that minimizing control is a linear function of the state,

$$u = -G x, G = R^{-1} B^T P \quad (2.20)$$

where P is obtained by solving the algebraic Riccati equation and is the only positive semi-definite matrix satisfying it,

$$A^T P + P A - P B R^{-1} B^T P + Q = 0 \quad (2.21)$$

Provided the weighting matrices meet the definiteness requirements stated above, the resulting loop is guaranteed to settle, which is one of the reasons for which the method was selected here in preference to pole placement, where the closed-loop pole locations must be chosen by the designer and the control effort does not enter the formulation.

The selection of Q and R constitutes the design decision, and the outcome depends upon it to a degree examined in Section 4.6. For a base-excited isolation problem the choice of the quantity to be penalised is not self-evident, since the states are relative displacements while the objective concerns absolute acceleration; this question is treated in Section 4.5.

2.9 NUMERICAL METHODS

All spectral quantities in this thesis were derived from Equation (2.15) evaluated line by line over the measured frequency content, so that no randomly generated input signal is required and the results are exactly repeatable. Where explicit time integration was needed for cross-checking, the embedded Runge–Kutta pair of Dormand and Prince [12] was employed, this being the method behind the ode45 solver.

The data acquisition unit produced its spectral estimates following Welch [13]. The procedure splits the record into segments that to some extent overlap, tapers each one, converts it, and then averages the separate estimates obtained, which smooth the estimate at the cost of frequency resolution. By Parseval’s theorem the total signal energy equals the sum of the squared spectral magnitudes, so the root-mean-square value follows directly from the spectrum without returning to the time record.

CHAPTER III

SYSTEM MODELLING

3.1 THE MACHINE AND ITS CABIN MOUNTING

The subject of the present work is a Hidromek HMK 480MG motor grader. The machine levels ground by means of a blade mounted between its axles and held in contact with the soil, advancing at a working speed of approximately 5 km/h while the operator maintains the blade at the required level.

The operator cabin is a welded steel structure carrying the glass panels and steering column, the seat, the instrument panel and the control levers. It rests on the chassis at eight mounting points; each fitted with an elastomeric mount assembly. Two mounting points lie forward, one on each side of the machine, and six are located to the rear in three pairs. The cabin, in the configuration in which it is delivered, has a mass of 919 kg.

The parameters required for the analysis were obtained from the manufacturer's engineering data and are listed in Table 1.

Table 1: Cabin Inertial Properties

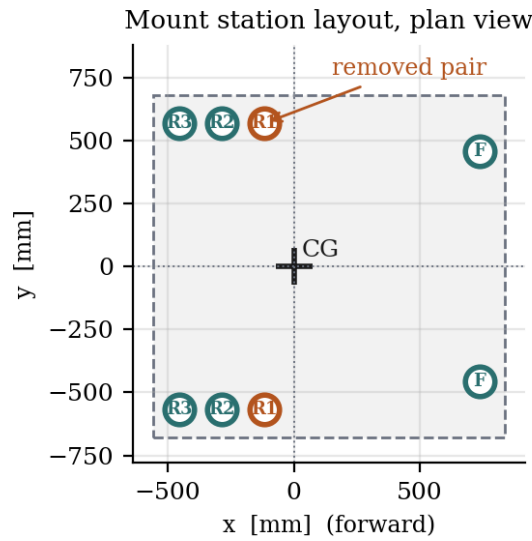
Property	Symbol	Value	Unit
Mass	m	919	kg
Pitch Moment of Inertia	I_{θ}	623	kg·m ²
Roll Moment of Inertia	I_{φ}	544	kg·m ²

The radius of gyration implied by these quantities are 0.823 m in pitch and 0.769 m in roll. Both are of the order of half the principal the overall cabin size, which is the dimension expected for a distributed structure of this size and provides a check on the internal consistency of the data among themselves.

Mounting positions are defined according to a right-handed coordinate system with its origin at the cabin's center of gravity, where the x-axis points forward along the machine, the y-axis points to the operator's right, and the z-axis points upwards. The coordinates are listed in Table 2.

Table 2: Mount Station Coordinates Relative to the Centre of Mass of the Cabin

Station	x [mm]	y [mm]
Front Right	+737.5	+456.6
Front Left	+737.5	-456.6
Rear Right 1	-116.3	+567.9
Rear Left 1	-116.3	-567.9
Rear Right 2	-286.3	+567.9
Rear Left 2	-286.3	-567.9
Rear Right 3	-456.3	+567.9
Rear Left 3	-456.3	-567.9

**Figure 2:** Mount Station Layout in Plan View and the Pair Marked R1 Lies Nearest the Centre of Gravity and Is the couple Taken out in the Six-Mount Configuration

Every station has a symmetrical counterpart in the plane $y = 0$, so the lateral coordinates sum exactly to zero. The longitudinal coordinates do not: their sum is -242.8 mm, locating the geometry centroid of the mount set 30.4 mm behind the Centre of gravity. This remaining unbalance becomes significant in Section 3.7.

3.2 THE MOUNT ASSEMBLY AND ITS LOAD PATH

Each unit consists of two elastomeric components installed in sequence. The upper component, designated mount rubber in the manufacturer's parts technical drawing, is an hollow-cylindrical element 70 mm tall with an outer diameter of 86 mm over its upper 22 mm and 53 mm over the remaining 48 mm. A steel bushing with an outer diameter of 40 mm and a length of 69 mm passes through the hole, while the fastener passes through the center of this bushing. The sub-component known as "Cushlon" is a smaller, cylindrical part positioned beneath the chassis bracket.

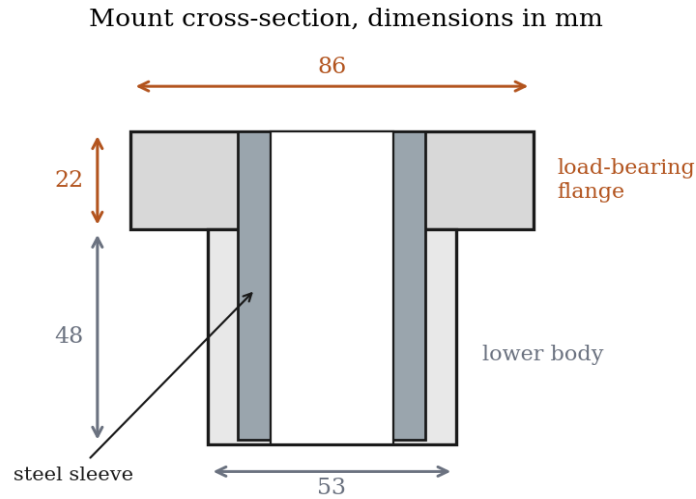


Figure 3: Cross-Section of the Mount Assembly Showing the Load-Bearing Flange, the Lower Body and the Steel Sleeve

Implementing which of the two carries the cabin weight is necessary before any stiffness can be assigned, and the answer follows from the assembly procedure. The cabin floor bears the upper face of the Mount Rubber flange. The lower face of that flange bears on the chassis bracket. The narrower lower body of the same component passes through the bracket aperture and locates the mount laterally. The Cushlon sits beneath the bracket and, with the washer and nut, closes the joint.

The vertical load therefore passes from the cabin, through the 22 mm flange, into the chassis bracket. The lower body of the Mount Rubber and the Cushlon lie outside this path: the former provides lateral location, and the latter completes the fastening, but neither is compressed by the cabin weight.

The steel sleeve is 1 mm shorter than the rubber column, so tightening the fastener pre-compresses the assembly by that amount. The distribution of this pre-compression between the flange and the lower body follows from their relative stiffnesses, which scale as the ratio of cross-sectional area to height. For the flange this ratio is 207 mm and for the lower body 20 mm, so the flange is 10.5 times the stiffer of the two. Elements in series share a given compression in inverse proportion to their stiffness, and the flange therefore absorbs 0.09 mm of the 1 mm total. This is negligible besides the deflection produced by the cabin weight, and the sleeve is accordingly excluded from the stiffness determination.

3.3 DETERMINATION OF MOUNT STIFFNESS

The stiffness of the load-bearing flange was obtained from its deflection under the cabin weight, measured on the assembled machine in the eight-mount configuration. The free height of the flange is 22.00 mm. Under load the height was measured at several stations and found to lie between 17.86 and 18.23 mm, giving deflections of 3.77 to 4.14 mm with a mean of 3.96 mm.

The static load carried by each station in the eight-mount arrangement is

$$W = m / n = 919 / 8 = 114.88 \text{ kgf} \quad (3.1)$$

and the static stiffness follows as

$$k_{static} = W / \delta = 114.88 / 3.96 = 29.05 \text{ kgf/mm} = 284\,936 \text{ N/m} \quad (3.2)$$

The spread in the measured deflection corresponds to a range of 27.75 to 30.47 kgf/mm, or ± 5 per cent about the adopted value.

Vibration analysis requires a dynamic rate. Applying the magnification factor of 1.3 stated on the component drawing,

$$k_{dyn} = 1.3 \times 29.05 = 37.76 \text{ kgf/mm} = 370\,426 \text{ N/m} \quad (3.3)$$

This value is used for every subsequent calculation. The static rate governs only the deflection under self-weight, reported in Section 3.9.

3.4 DETERMINATION OF THE DAMPING

No measurement of the mount damping was available. Neither the force–deflection hysteresis loop nor a free-decay record was obtained, and the component drawing states no loss factor. The damping is accordingly the one parameter of the model that had to be determined by other means.

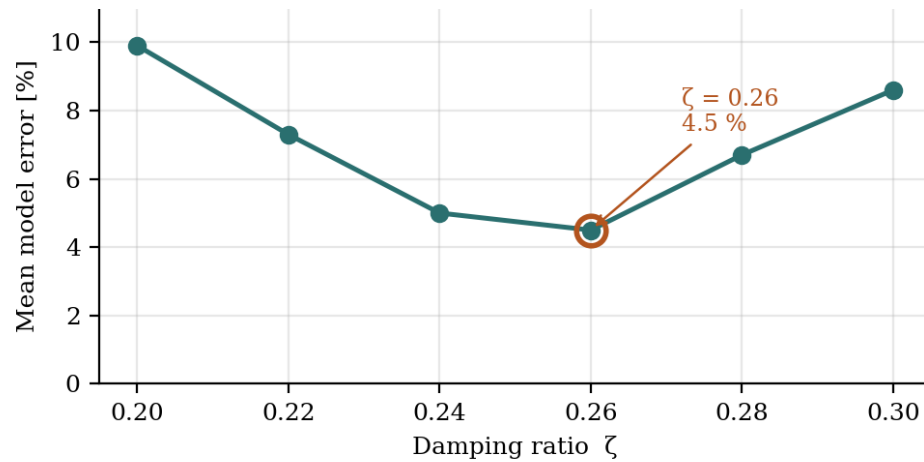
The first approach considered was to adopt a value from literature. Loss factors between 0.05 and 0.15 are quoted for the elastomers used in cabin mounts, corresponding to damping ratios of 0.025 to 0.075. Table 3 shows the cabin acceleration predicted by the model for values in this range, compared with the measured value. The procedure by which the prediction is formed is described in Section 3.10; it is stated here only that the measured chassis spectrum is passed through the model and the resulting cabin response computed.

Table 3: Model Error for Damping Values Taken from the Literature

Loss Factor η	Damping Ratio ζ	Mean Model Error
0.05	0.025	209 %
0.10	0.050	92 %
0.15	0.075	55 %

Even at the upper end of the quoted range the model overpredicts the cabin response by a factor exceeding three. A lightly damp system produces a resonance peak so pronounced that the predicted response bears no relation to the measurement. Literature values were therefore rejected.

The damping was instead identified from the measured response. For each candidate value the model prediction was formed for the three operating conditions of the eight-mount configuration, and the mean deviation from the measurement computed. Table 4 records the sweep. Mean model error against the damping ratio. The minimum defines the adopted value.

**Figure 4:** Mean Model Error Against the Damping Ratio; The Minimum Defines the Adopted Value**Table 4:** Identification of the Damping Ratio

ζ	Coarse Grading	Road Opening	Fine Grading	Mean
0.20	8.2 %	7.3 %	14.2 %	9.9 %
0.22	9.4 %	4.1 %	8.5 %	7.3 %
0.24	10.5 %	1.1 %	3.6 %	5.0 %
0.26	11.4 %	1.4 %	0.5 %	4.5 %
0.28	12.3 %	3.6 %	4.1 %	6.7 %
0.30	13.0 %	5.5 %	7.3 %	8.6 %

The minimum occurs at $\zeta = 0.26$, and this value is adopted. It corresponds to a loss factor of 0.52, which lies above the range quoted for new rubber components. Two

explanations are consistent with the observation. The mounts on the machine have been in service, and rubber stiffens and gains internal friction as it ages. Alternatively, the fitted value may be absorbing model error that is not physically damping. The question is revisited in Chapter VI.

The damping ratio is a property of the configuration rather than of the component, since it depends on the total stiffness and hence on the number of stations. The quantity carried forward is the damping coefficient per station, obtained from the definition of the damping ratio applied to the eight-mount arrangement,

$$C_{11} = 2 \zeta \sqrt{K_{11} m} = 2 \times 0.26 \times \sqrt{2\,963\,405 \times 919} = 27\,136 \, N \cdot s/m \quad (3.4)$$

$$c = C_{11} / n = 27\,136 / 8 = 3392 \, N \cdot s/m \quad (3.5)$$

This coefficient describes the mount itself and applies unchanged to any number of stations.

Table 5: Mount Properties

Property	Symbol	Value	Unit
Static Stiffness Per Station	k_{static}	29.05	kgf/mm
Dynamic Stiffness Per Station	k_{dyn}	37.76	kgf/mm
Damping Coefficient Per Station	c	3392	N·s/m
Damping Ratio, Eight Mounts	ζ	0.26	—

3.5 EQUATION OF MOTION

3.5.1 Coordinates

The cabin is represented as a rigid body with three degrees of freedom, collected in

$$q = [z, \theta, \varphi]^T \quad (3.6)$$

where z is the vertical displacement of the Centre of gravity relative to the chassis, θ the pitch rotation about the transverse axis and φ the roll rotation about the longitudinal axis.

The choice of relatives rather than absolute coordinates is dictated by the mounts. A mount develops force in response to its own deformation, which is the difference between the motion of its two ends. Writing the equations in relative coordinates makes the mount forces functions of the coordinates directly.

3.5.2 Mount Deformation

For small rotations the vertical deformation of the mount at station i , located at (x_i, y_i) , is

$$d_i = z + \theta x_i + \varphi y_i = v_i^T q, v_i = [1, x_i, y_i]^T \quad (3.7)$$

The vector v_i is termed the participation vector of station i .

3.5.3 Force and Moment Balance

By Equation (2.16) the force developed at station i is

$$F_i = k d_i + c \dot{d}_i = k v_i^T q + c v_i^T \dot{q} \quad (3.8)$$

acting to oppose the deformation. Summing the vertical forces and the moments about the two horizontal axes through the Centre of gravity gives three equations, which in matrix form are

$$M \ddot{q} + C \dot{q} + K q = f \quad (3.9)$$

with

$$M = \text{diag}(m, I_\theta, I_\varphi) \quad (3.10)$$

$$K = k \sum_i v_i v_i^T, C = c \sum_i v_i v_i^T \quad (3.11)$$

Both matrices are formed from the same geometric construction and differ only in the multiplier, a property termed proportional damping whose consequence is examined in Section 3.8.

3.5.4 Base Excitation

The chassis motion enters as a base excitation. Let z_b denote the absolute vertical displacement of the chassis and Z that of the cabin, so that $z = Z - z_b$. Newton's second law applies to absolute acceleration,

$$m \ddot{Z} = -(\text{forces from the mounts}) \quad (3.12)$$

Substituting $\ddot{Z} = \ddot{z} + \ddot{z}_b$ and transferring the base term to the right-hand side,

$$m \ddot{z} + c \dot{z} + k z = -m \ddot{z}_b \quad (3.13)$$

The term on the right is not a physical force. It arises from the transformation to the relative coordinate and expresses the inertial effect of describing the motion in a frame that is itself accelerating. Extended to three degrees of freedom,

$$M \ddot{q} + C \dot{q} + K q = -M [1, 0, 0]^T \ddot{z}_b + B_u u \quad (3.14)$$

in which $B_u u$ represents the actuator forces introduced in Chapter IV and is zero for the passive system.

3.5.5 Expanded Form

Writing Equation (3.14) in full,

$$m \ddot{z} + C_{11} \dot{z} + C_{12} \dot{\theta} + C_{13} \dot{\varphi} + K_{11} z + K_{12} \theta + K_{13} \varphi = -m \ddot{z}_b \quad (3.15)$$

$$I_\theta \ddot{\theta} + C_{21} \dot{z} + C_{22} \dot{\theta} + C_{23} \dot{\varphi} + K_{21} z + K_{22} \theta + K_{23} \varphi = 0 \quad (3.16)$$

$$I_\varphi \ddot{\varphi} + C_{31} \dot{z} + C_{32} \dot{\theta} + C_{33} \dot{\varphi} + K_{31} z + K_{32} \theta + K_{33} \varphi = 0 \quad (3.17)$$

Only the first equation carries the base excitation, since the chassis motion considered is purely vertical. The rotational degrees of freedom are excited only through the coupling terms in K and C.

3.6 SYSTEM MATRICES

Carrying out the summation in Equation (3.11) and writing the result in terms of the geometric sums of the mount coordinates,

$$K = k \cdot [n, \Sigma x_i, \Sigma y_i ; \Sigma x_i, \Sigma x_i^2, \Sigma x_i y_i ; \Sigma y_i, \Sigma x_i y_i, \Sigma y_i^2] \quad (3.18)$$

Each entry is the dynamic stiffness multiplied by one geometric sum. Table 6 gives these sums for the eight-mount configuration and for the six-mount configuration whose selection is treated in Section 3.7.

Table 6: Geometric Sums of the Mount Coordinates

Sum	Eight Mounts	Six Mounts	Unit
n	8	6	—
Σx_i	-0.2428	-0.0102	m
Σy_i	0	0	m
Σx_i^2	1.6952	1.6682	m ²
Σy_i^2	2.3520	1.7070	m ²
$\Sigma x_i y_i$	0	0	m ²

Two entries canceled completely. The lateral symmetry of the mount layout makes Σy_i and $\Sigma x_i y_i$ zero term by term, and accordingly, K_{13} and K_{23} are identically zero. Roll is therefore decoupled from heave and pitch, and a purely vertical base excitation cannot excite it under any condition.

The two actuator commands are identical, since purely vertical input does not excite roll.

The resulting matrices for the six-mount configuration are

$$K = \begin{bmatrix} 2\,222\,554 & -3778 & 0 \\ -3778 & 617\,932 & 0 \\ 0 & 0 & 632\,320 \end{bmatrix} \quad (3.19)$$

$$C = \begin{bmatrix} 20\,353 & -35 & 0 \\ -35 & 5659 & 0 \\ 0 & 0 & 5790 \end{bmatrix} N \cdot s/m \quad (3.20)$$

Table 7 lists the principal entries for both configurations.

Table 7: Lists the Principal Items for Both Configurations

Entry	Eight Mounts	Six Mounts	Unit
K_{11}	2 963 405	2 222 554	N/m
K_{12}	-89 939	-3778	N/rad
K_{22}	627 952	617 932	N·m/rad
K_{33}	871 252	632 320	N·m/rad

The heave–pitch coupling term demands close examination. Expressed as a fraction of the leading diagonal item, it falls from 3.03 per cent with eight mounts to 0.17 per cent with six, so that in the six-mount configuration the vertical and pitching motions are close to entirely independent.

3.7 SELECTION OF THE MOUNT CONFIGURATION

3.7.1 The Effect of Mount Count

Extraction of the mounts reduces the total stiffness in fraction to the number removed, and the natural frequency concurrently decreases as the square root of that number. Table 8 sets out the implications.

Table 8: Effect of the Number of Mount Stations

n	K_{11} [N/m]	f_n [Hz]	Threshold $\sqrt{2f_n}$ [Hz]	Load per station [kgf]	Utilization
8	2 963 405	9.04	12.79	114.9	38 %
6	2 222 554	7.83	11.07	153.2	51 %
4	1 481 703	6.39	9.04	229.8	77 %
3	1 111 277	5.53	7.83	306.3	102 %

The allowable compressive load stated for the mount is 300 kgf. Three mounts exceed it and four leave a margin of 23 per cent, which was deemed inadequate for a component that must also resist braking forces, cornering and slope

operation. Six mounts retain a margin of 49 per cent against the allowable value and a factor of 6.5 against the proof load of 1000 kgf.

3.7.2 Which Pair to Remove

Three rear pairs are accessible for extraction, and the vertical stiffness of the remaining set is identical in each case, since K_{11} depends only on the number of mounts. The pairs are characterized by their effect on the longitudinal balance.

Table 9: Consequence of Removing Each Rear Pair

Pair Removed	x [mm]	Σx_i Remaining [mm]	K_{12} [N/rad]
Rear 1	-116.3	-10.2	-3778
Rear 2	-286.3	+329.8	+122 167
Rear 3	-456.3	+669.8	+248 111

Removing the pair closest the Centre of gravity reduces the residual imbalance from -242.8 mm to -10.2 mm. Either of the other two pairs increases the imbalance and inverts its sign, producing a coupling term larger than that of the original eight-mount arrangement. The first pair is selected accordingly.

The choice has a further consequence for Chapter IV. The two vacated positions lie at $x = -116.3$ mm, and it will be shown that placement of actuators there generates a induced pitch moment.

3.8 MODAL PROPERTIES

Solving the eigenvalue problem of Equation (2.9) for the six mount configuration yields the experimental data in Table 10.

Table 10: Modal Properties, Six-Mount Configuration

Mode	ω^2 [s ⁻²]	f [Hz]	ζ	Mode shape $[z, \theta, \phi]$	Dominant motion
1	991.8	5.01	0.144	[+0.003, +1.000, 0]	Pitch
2	1162.4	5.43	0.156	[0, 0, +1.000]	Roll
3	2418.5	7.83	0.225	[+1.000, -0.004, 0]	Heave

Three modes of solution warrant consideration

The second mode is exactly decoupled: its heave and pitch components are zero rather than only small; a direct consequence of the diminishing geometric sums identified in Section 3.6.

The remaining two modes are loosely coupled. The heave mode contains 0.4 per cent pitch participation, so that each mode may be established with a single coordinate. As a check, computing each spectrum from its own diagonal entry alone gives 7.83, 5.01 and 5.43 Hz, agreeing with the eigenvalue solution to within the precision referenced.

The modal damping ratios vary across modes. Because C is proportional to K , the damping ratio scales with frequency, rising from 0.144 in pitch to 0.225 in heave. The heave value governs the vertical response and is the quantity referred to in the remainder of this work.

3.9 STATIC DEFLECTION

The static deflection follows from the static stiffness and the load per station,

$$\delta = m g / (n k_{static}) \quad (3.21)$$

For six stations this gives 5.27 mm, against 3.95 mm for eight. Both are of the order of a fifth of the free height of the load-bearing flange, which is consistent with the elastic range of the material.

The deflection is the quantity that limits any further softening of the passive system. Combining Equations (2.2) and (3.21) gives δ proportional to $1/f_n^2$, so reducing the resonance from 7.83 Hz to 5.5 Hz would require a deflection of 10.7 mm, twice the present value. Chapter IV shows that active control achieves an equivalent reduction in dynamic stiffness with no change in deflection whatsoever.

3.10 MEASUREMENT CAMPAIGN

3.10.1 Instrumentation and Procedure

Vibration was recorded at four points on the machine: on the cabin floor at the left and right sides, and on the chassis on the mounting plane on the same two sides. Triaxial accelerometers were used at every point, so that twelve channels were acquired simultaneously.

The acquisition rate was 1024 Hz, giving spectra of 512 lines at 1 Hz resolution over the range 0 to 511 Hz. Each condition was recorded for 120 seconds. The engine was held at 1950 rpm and the machine travelled at approximately 5 km/h, tyre pressures having been set to 40 psi before the tests and the machine brought to operating temperature.

Three ground-engaging conditions were recorded, listed in Table 11.

Table 11: Operating Conditions

Condition	Description
Coarse Grading	Blade engaged, heavy cut
Road opening	Blade engaged, first pass on unprepared ground
Fine grading	Blade engaged, light finishing cut

The campaign was then repeated with the two mount stations at $x = -116.3$ mm removed, on the same day, with the same operator, the same machine settings and the same section of ground. This procedure is the reason the comparison reported in Chapter V may be attributed to the mounting change alone.

3.10.2 Data Reduction

The two sensors on each side measure the same quantity at two points. They are combined by averaging their energies rather than their amplitudes,

$$S(f) = \sqrt{([S_L^2(f) + S_R(f)^2] / 2)} \quad (3.22)$$

The root-mean-square value of each channel follows from Parseval's theorem as the square root of the sum of the squared spectral amplitudes,

$$a_{rms} = \sqrt{\sum_f S(f)^2} \quad (3.23)$$

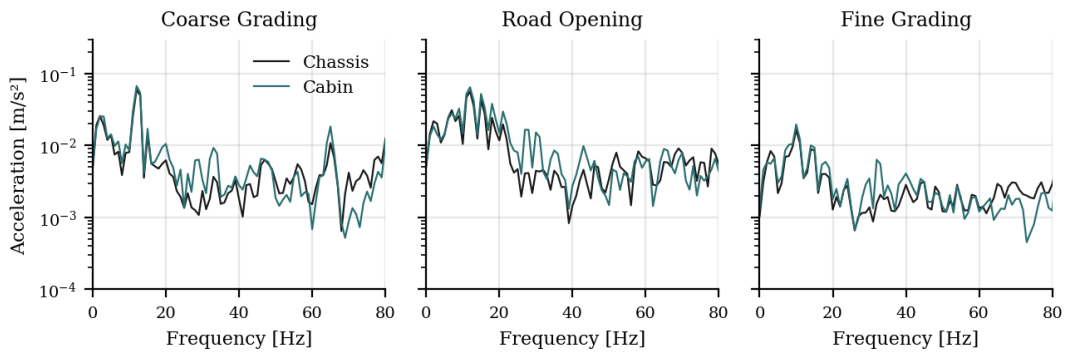


Figure 5: Measured Vertical Acceleration Spectra for the Eight-Mount Configuration. The Chassis Carries Substantive Data Above 100 Hz from the Engine, While the Cabin Response Is Concentrated Below 40 Hz

3.10.3 Measurement Quality

The ratio of the right-hand to the left-hand sensor reading provides a check on the acquisition, since the two should be comparable under conditions that do not favor

one side. The ratio lay between 0.92 and 1.29 in every condition and configuration, confirming that neither sensor drifted during the campaign.

3.10.4 Measured Results

Table 12: Measured Vertical Acceleration and Transmissibility

Condition	<i>n</i>	Chassis [m/s ²]	Cabin [m/s ²]	<i>T</i>
Coarse grading	8	0.1759	0.1239	0.705
	6	0.1920	0.1223	0.637
Road opening	8	0.2112	0.1590	0.753
	6	0.2216	0.1470	0.663
Fine grading	8	0.1212	0.0498	0.411
	6	0.1594	0.0637	0.400
Mean	8	0.1694	0.1109	0.623
	6	0.1910	0.1110	0.567

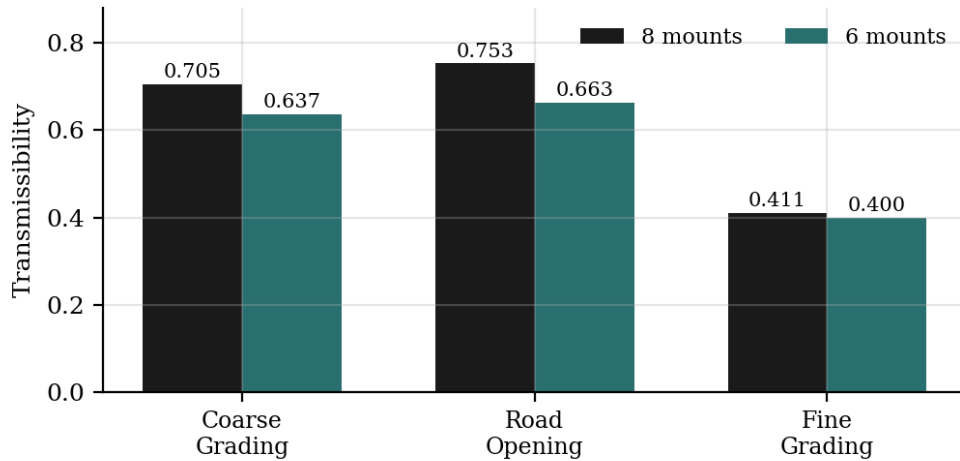


Figure 6: Measured Transmissibility by Condition and Configuration

Transmissibility is lower in the six-mount configuration in every condition. The absolute cabin acceleration is nearly identical between the two, but the chassis input was 13 per cent higher during the six-mount campaign, so the transmissibility rather than the absolute value is the measure of isolation performance.

Measurements on all three axes, given in Appendix 2, show the cabin response to be largest in the vertical direction under every condition, which supports the selection of vertical acceleration as the control objective.

3.11 EXCITATION FREQUENCY

The dominant excitation frequency was obtained from the measured chassis spectrum as its energy-weighted centroid,

$$f_{exc} = \frac{\sum_f f S(f)^2}{\sum_f S(f)^2} \quad (3.24)$$

The centroid is preferred to the peak location because it accounts for the entire spectrum and is consequently less sensitive to noise on any single line.

The summation is taken over 2 to 60 Hz. The lower bound excludes sensor drift and low-frequency noise. The upper limit excludes engine-induced content: the firing frequency of the six-cylinder engine appears around 75 Hz, and this contribution is not part of the ground-borne vibration the mounts are designed to isolate. Raising the frequency range to 80 Hz shifts the center of gravity upward by approximately 3 Hz without providing any additional information regarding ground interaction.

Table 13: Excitation Frequency by Condition

Condition	Eight mounts	Six mounts
Coarse grading	12.23	24.93
Road opening	13.30	14.02
Fine grading	13.64	15.07

Excluded.

This value diverges from the other five by a margin approaching a factor of two. Excitation is a characteristic of ground contact and cannot depend on the number of connecting elements beneath the cabin; therefore, the deviation in question stems from the measurement rather than the machine itself.

The meaning of the five remaining values is

$$f_{exc} = 13.65 \text{ Hz} \quad (3.25)$$

with a standard deviation of 0.93 Hz. The narrowness of this range across the two configurations and three conditions supports the interpretation that arousal is a characteristic of the process.

3.12 OPERATING POINT

Combining the excitation frequency with the natural frequencies yields the frequency ratio at which the system operates.

Table 14: Operating Point

Configuration	f_n [Hz]	Threshold $\sqrt{2f_n}$ [Hz]	$r = f_{exc}/f_n$	Margin above $\sqrt{2}$
Eight Mounts	9.04	12.79	1.51	6.8 %
Six Mounts	7.83	11.07	1.74	23.3 %

Since both configurations operate within the isolation zone, the fasteners dampen vibration rather than amplifying it. However, the eight-fastener arrangement exceeds the threshold value by less than seven percent. Removing two stations increases this margin to twenty-three percent, which represents the quantitative expression of the improvement specified in Section V.

3.13 MODEL VALIDATION

3.13.1 Procedure

The model is validated by feeding it the measured chassis spectrum and comparing its predicted cabin acceleration with the measured cabin acceleration. The cabin measurement is not used as an input at any stage; it serves solely as a reference against which the prediction is tested.

For each frequency line the transmissibility of Equation (2.5) is evaluated at $r = f/f_n$, multiplied by the measured chassis amplitude at that line, and the results combined,

$$a_{cabin} = \sqrt{\sum_f [|H(f)|S_{ch}(f)]^2} \quad (3.26)$$

No single-frequency assumption enters this calculation. The entire measured spectrum is used, so the procedure remains valid whether the excitation is narrow-band or broadband.

3.13.2 Results

Table 15: Model Validation, Eight-Mount Configuration

Condition	Model [m/s ²]	Measured [m/s ²]	Error
Coarse Grading	0.1098	0.1239	11.4 %
Road Opening	0.1568	0.1590	1.4 %
Fine Grading	0.0495	0.0498	0.5 %
Mean			4.5 %

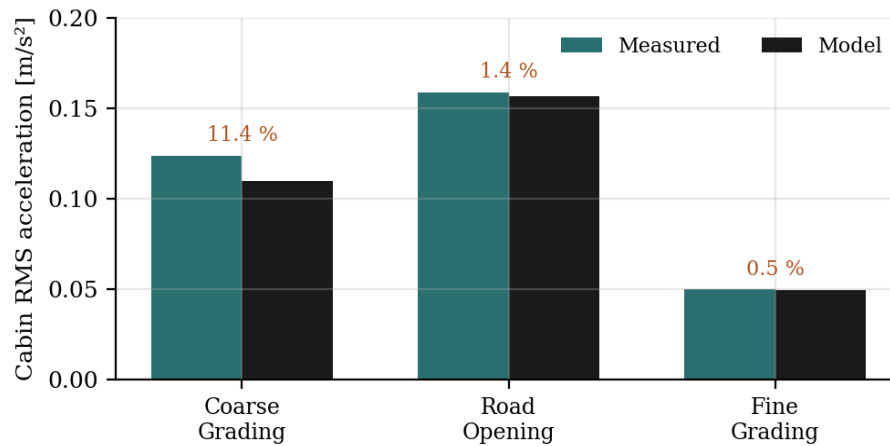


Figure 7: Model Validation Against the Measured Cabin Acceleration, Eight-Mount Configuration. Percentages Give the Deviation

The mean error of 4.5 per cent is obtained with a single fitted parameter tested against three independent conditions, leaving two degrees of freedom in the comparison. Had the structure of the model been incorrect, no single damping value could have reproduced three conditions whose chassis amplitudes differ by a factor of 1.7.

3.13.3 The Six-Mount Prediction

The damping was identified in the eight-mount configuration, and that configuration therefore constitutes the validation. The six-mount response is a forward prediction from the same parameters, and Table 16 compares it with the measurement.

Table 16: Six-Mount Prediction

Condition	Model [m/s²]	Measured [m/s²]	Deviation
Coarse Grading	0.0749	0.1223	38.7 %
Road Opening	0.1204	0.1470	18.1 %
Fine Grading	0.0447	0.0637	29.8 %
Mean			28.9 %

The model underpredicts the measured response in all three conditions, and the deviation is systematic rather than scattered. The most probable explanation is that the damping does not remain constant when the load per station rises from 114.9 to 153.2 kgf. The dynamic properties of filled rubber depend on the mean strain, and the model treats the damping coefficient as a fixed quantity.

The consequence for the results of Chapter V is that the control performance computed on the six-mount model should be regarded as an upper estimate. This is stated explicitly among the limitations in Chapter V

CHAPTER IV

ACTIVE VIBRATION CONTROL DESIGN

4.1 THE PRINCIPLE OF ACTIVE ISOLATION

The constraint identified in Chapter III has a single origin: one stiffness value must serve two purposes that require opposite properties. Carrying the cabin weight without excessive deflection calls for a high value; placing the resonance below the excitation band calls for a low one.

An actuator placed in the load path separates the two functions. The passive mounts continue to carry the static weight, since the actuator produces no force when the cabin is at rest. When the cabin moves relative to the chassis, the actuator generates a force opposing the dynamic component of the mount reaction. The stiffness experienced under vibration is thereby reduced while the stiffness felt under the static weight remains the same.

Expressing the control force as a fraction α of the force passed on by the mounts, the vertical equation of motion takes the form

$$m \ddot{z} = -(1 - \alpha) (K q + C \dot{q}) \quad (4.1)$$

The effective stiffness and the effective damping are both multiplied by $(1 - \alpha)$, and the effective natural frequency follows as

$$f_{eff} = f_n \sqrt{1 - \alpha} \quad (4.2)$$

For $\alpha = 0.90$ the effective stiffness is reduced to a tenth of its nominal value and the resonance from 7.83 Hz to 2.48 Hz. Table 17 sets shows the consequences.

Table 17: Effect of Control Action, Six-Mount Configuration

Quantity	Passive	$\alpha = 0.90$
Effective stiffness	100 %	10 %
Heave resonance	7.83 Hz	2.48 Hz
Frequency ratio at 13.65 Hz	1.74	5.50
Static deflection	5.27 mm	5.27 mm

The final row is the key result. Obtaining a resonance of 2.48 Hz by passive means would require a static deflection of about 52 mm, since the deflection is inversely proportional to with the square of the natural frequency. The actuator responds to motion and not to load, so the deflection is independent of it. The trade-off that constrains passive design does not apply to this problem.

4.2 CONFIGURATIONS EXAMINED

Two hardware configurations were evaluated.

Configuration A takes the place of the two removed mount stations with actuators, so that six passive mounts and two actuators support the cabin. This is the configuration adopted and is used for all results reported in Chapter V.

Configuration B Protects all eight connection points and adds two actuators. It does not require any assembly changes and, consequently, a re-evaluation of the load per station, but it forgoes the passive improvement derived from the six-mount arrangement.

Configuration A was adopted because it combines both sources of improvement, and because the load factor of 51 per cent established in Section 3.7 leaves an acceptable margin.

4.3 ACTUATOR PLACEMENT

4.3.1 The Parasitic Pitch Moment

The two positions freed by the removed mounts lie at $x = -116.3$ mm. Applying the actuators there appears natural, since the mounting points already exist, but it introduces a difficulty.

Both actuators generate vertical force. When commanded together to oppose a downward cabin motion, they generate an upward force at a station 116.3 mm behind the Centre of gravity, and therefore also a pitching moment

$$M_{\theta} = 2 F |x_a| \quad (4.3)$$

For the design force these amounts to approximately 60 Nm. The consequence is that the controller, in acting on the vertical motion, excites the pitch motion. The pitch response grows instead of falling.

4.3.2 Alignment with the Centre of Gravity

Positioning both actuators at the longitudinal station of the centre of gravity, $x_a = 0$, reduces the moment arm to zero and eliminates the effect. The actuators retain their lateral separation at $y_a = \pm 567.9 \text{ mm}$, in line with mount stations.

The actuator distribution matrix follows from these positions,

$$B_u = [1, 1 ; 0, 0 ; 0.5679, -0.5679] \quad (4.4)$$

The rows correspond to the vertical, pitch and roll equations respectively. The zero second row is the algebraic statement of the placement decision: the actuators cannot provide a pitch moment.

Two modes of operation follow. Identical commands to both units produce pure vertical force with no moment of either axis. Identical and opposite commands produce pure roll moments with no net vertical force. Two actuators therefore supply independent authority over two degrees of freedom.

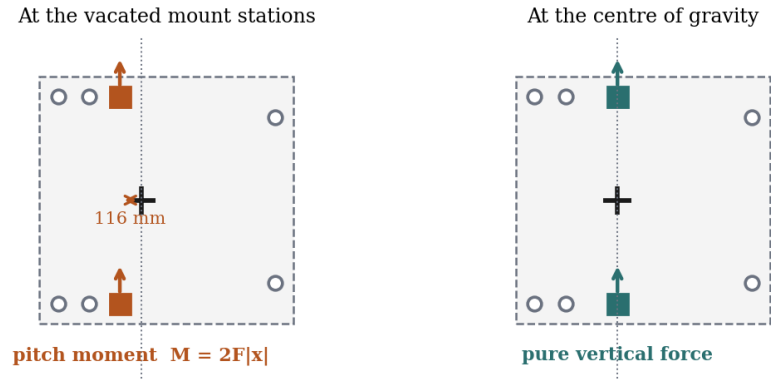


Figure 8: Actuator Coordinate. Positioning At the Vacated Mount Stations Produces a Pitch Moment; Alignment with the Centre of Gravity Gives Pure Vertical Force

4.3.3 Consequence for Pitch

Pitch is uncontrollable under this arrangement. The consequence is slight in the present application: the base excitation is purely vertical, and Section 3.6 established that the heave–pitch coupling term in the six-mount configuration amounts to 0.17 per cent of the leading diagonal entry. Pitch is therefore barely exciting, and little is lost by placing it beyond the reach of the controller.

4.4 PID-BASED CONTROL

4.4.1 The Control Law

The law is constructed to cancel a defined fraction of the force the mounts transmit into the cabin. The vertical mount reaction is given by the first row of the stiffness and damping matrices acting on the state, so that each of the two actuators is commanded

$$u_1 = u_2 = (\alpha / 2)(K_{(1, \cdot)} q + C_{(1, \cdot)} \dot{q}) \quad (4.5)$$

The division by two distributes the required force equally between the units. Two measurements are needed: the relative displacement and the relative velocity of the cabin with respect to the chassis.

The law is proportional derivative in structure, the first term acting on displacement and the second on velocity. Section 4.4.2 establishes the value of the integral gain.

4.4.2 Determination of the Integral Gain

The general PID law contains three gains,

$$u = K_p e + K_i \int e dt + K_d \dot{e} \quad (4.6)$$

The proportional and derivative gains follow directly from the stiffness and damping matrices as shown in Equation (4.5). The integral gain is determined by examining what it would contribute.

The integral term exists to remove a persistent error that the proportional action cannot eliminate. Three considerations show that no such error arises here.

The closed loop retains positive stiffness for any α below unity, as Equation (4.1) shows. The system therefore returns to its equilibrium position of its own accord after a disturbance, and no steady state offset develops for the integral term to accumulate.

More significantly, the weight of the cabin holds the mounts in permanent compression. The integral term interprets this continuous deviation as an error requiring correction and, over time, begins to attenuate the static support force. The cabin will gradually sag downwards. The entire design relies on the principle that the connecting elements continue to bear the static load; the integral term, however, disrupts precisely this mechanism.

The integral action also introduces phase lag at low frequency, where the passive mounts already perform well. It would therefore reduce the stability margin without any compensating benefit.

The integral gain is accordingly determined to be zero,

$$K_i = 0 \quad (4.7)$$

The law reduces to proportional–derivative form. This is the outcome of the design rather than an omission from it: the integral gain has been evaluated and found to be appropriately zero for this class of system.

4.4.3 Selection of the Cancellation Fraction

The fraction α is the single tuning parameter. Table 18 gives the performance obtained across the admissible range, computed by the procedure of Section 4.7.

Table 18: Effect of the Cancellation Fraction

α	K_{eff}/K	f_{eff} [Hz]	Reduction	Actuator Force [kN]
0.30	0.70	6.55	6.3 %	0.03
0.50	0.50	5.53	6.5 %	0.06
0.70	0.30	4.29	14.0 %	0.13
0.80	0.20	3.50	23.5 %	0.19
0.90	0.10	2.48	45.6 %	0.28
0.95	0.05	1.75	59.7 %	0.36

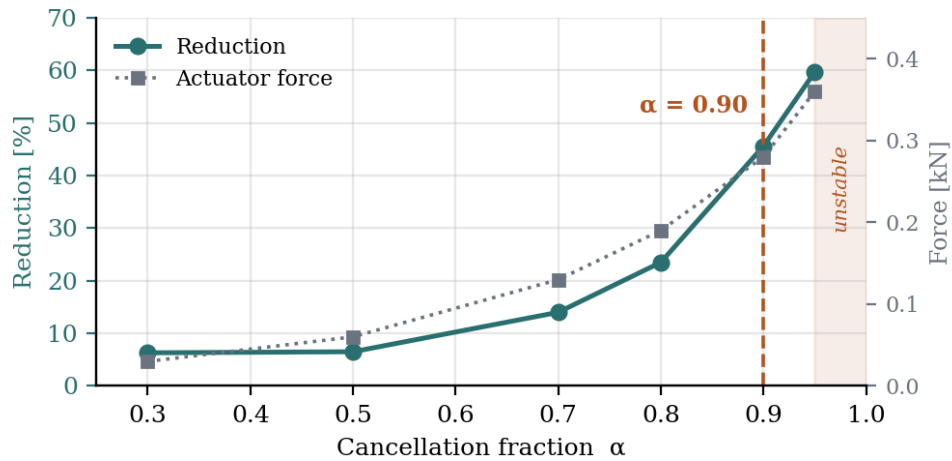


Figure 9: Reduction and Actuator Force Against the Cancellation Fraction

Performance improves monotonically with α , which raises the question of why the largest value is not adopted. Three considerations limit it.

At $\alpha = 1$ the effective stiffness vanishes entirely, and the cabin is no longer restrained. It would float free, with no mechanism to return it to its equilibrium position under braking, cornering or slope operation.

Above unity the effective stiffness becomes negative. A cabin displaced from equilibrium would then be pushed further from it rather than restored, and the loop would be unstable.

The third consideration is decisive. The control law computes the mount reaction from the modelled stiffness and damping matrices. Should the actual mounts be stiffer than modelled by twenty per cent, the controller would cancel more than the commanded fraction, and the effective α would exceed the value set. A choice of 0.90 leaves a margin of ten per cent for such an error, whereas 0.98 would leave two per cent and a modest modelling error would carry the system past unity.

The value (4.8) $\alpha = 0.90$ is therefore adopted. It forgoes fourteen percentage points of reduction in exchange for a margin against modelling error and is a deliberate safety decision rather than a performance optimum.

4.5 LINEAR QUADRATIC REGULATOR

4.5.1 Formulation of the Objective

The regulator determines the feedback gain by minimizing a quadratic cost functional. The construction of that functional is the design decision, and in a base-excited isolation problem it is not self-evident.

The state variables of the model is relative displacements. The intuitive choice would be to penalise them, on the reasoning that the cabin should move as little as possible. This reasoning is incorrect, and the reason is instructive.

Consider what good isolation looks like above resonance. The chassis moves up and down; the cabin remains where it is; the operator feels nothing. In this state the distance between cabin and chassis changes continuously, so the relative displacement is large. A large relative displacement is therefore a signature of good isolation, not a defect.

Penalising it would drive it towards zero. Zero relative displacement means the cabin follows the chassis exactly, which is the definition of no isolation at all. The regulator would use the actuators to lock the cabin to the chassis, producing a result worse than applying no control.

The objective is accordingly constructed on the quantity the operator actually experiences, namely the absolute vertical acceleration of the cabin,

$$J = \int_0^\infty (\ddot{z}_{cabin}^2 + \rho_u u^T u) dt \quad (4.9)$$

The first term penalises what is felt. The second penalises the actuator effort, so that the force required enters the design rather than emerging from it. The weight ρ_u sets the balance between the two and is determined in Section 4.6.

4.5.2 Expression of the Output

Absolute cabin acceleration is the sum of the relative and base accelerations,

$$\ddot{z}_{cabin} = \ddot{z} + \ddot{z}_b \quad (4.10)$$

Taking the first row of the equation of motion and substituting,

$$\ddot{z}_{cabin} = -(1/m)(K_{(1, \cdot)} q + C_{(1, \cdot)} \dot{q}) + (1/m)(u_1 + u_2) \quad (4.11)$$

The base acceleration cancels. Once the actuator forces are carried as states, as Section 4.5.3 arranges, the absolute acceleration becomes a function of the state alone,

$$\ddot{z}_{cabin} = C_z x \quad (4.12)$$

and the problem assumes the standard form with

$$Q = C_z^T C_z, R = \rho_u I \quad (4.13)$$

4.5.3 Actuator Dynamics

A hydraulic actuator cannot realise a commanded force instantaneously. Representing the response by a first-order lag of bandwidth f_a ,

$$\dot{F}_i = \omega_a (F_{cmd, i} - F_i), \omega_a = 2\pi f_a \quad (4.14)$$

the actuator forces become state variables in their own right. The state vector accordingly has eight elements,

$$x = [z, \theta, \varphi, \dot{z}, \dot{\theta}, \dot{\varphi}, F_1, F_2]^T \quad (4.15)$$

and the system matrices are augmented correspondingly. A bandwidth of 40 Hz was adopted, representative of commercial servo-hydraulic units of the required force capacity.

Including the actuator dynamics rather than assuming instantaneous response has two consequences. It prevents the performance from being overstated, and it allows the regulator to anticipate the lag: because the actual force is a state, the controller knows how far the actuator has progressed towards its command and can act

accordingly. This is one reason for the margin by which the regulator outperforms the proportional–derivative law, which responds only to the instantaneous measurement.

4.5.4 Solution

With A, B, Q and R established, the algebraic Riccati equation

$$A^T P + P A - P B R^{-1} B^T P + Q = 0 \quad (4.16)$$

is solved for P, and the gain follows as

$$G = R^{-1} B^T P \quad (4.17)$$

The gain matrix has dimensions 2×8 : it accepts the eight states and produces the two force commands. The control law is

$$u = -G x \quad (4.18)$$

Both controllers therefore reduce to the same structural form. The difference lies entirely in how the gain is obtained: from the mount properties directly in the case of the PID-based law, and from the solution of Equation (4.16) in the case of the regulator. In closed loop the system matrix A is replaced by $A - BG$ in both cases.

4.6 SELECTION OF THE CONTROL WEIGHT

The weight ρ_u is the sole design parameter of the regulator, and the performance obtained depends on it to a degree that merits explicit treatment. Table 19 records a sweep across five decades.

Table 19: Effect of the Control Weight

ρ_u	Reduction	Actuator Force [kN]	Dominant Closed-Loop Pole
1×10^{-8}	52.5 %	0.24	−4.54
1×10^{-9}	67.1 %	0.48	−4.54
1×10^{-10}	79.3 %	0.73	−2.80
1×10^{-11}	88.1 %	1.16	−1.57
1×10^{-12}	93.0 %	3.00	−0.89

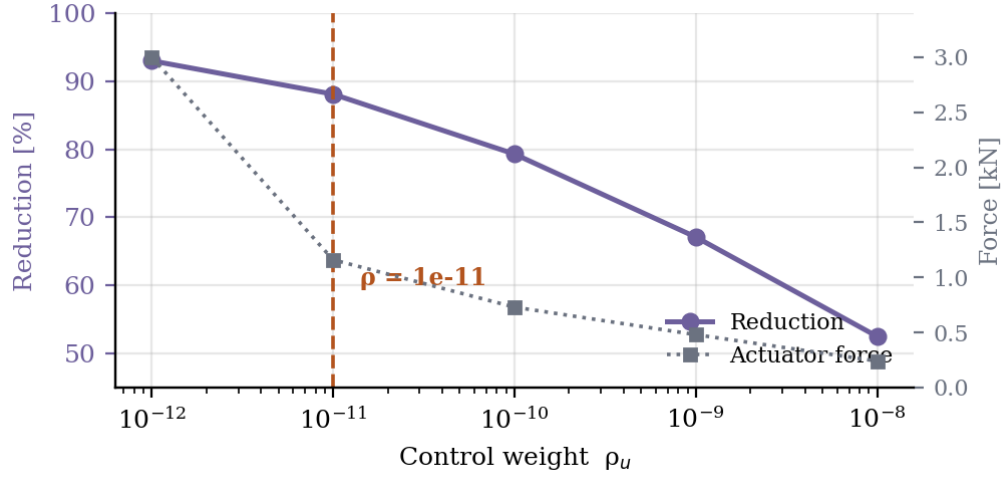


Figure 10: Reduction and Actuator Force Against the Control Weight, Swept Across Five Decades

Two observations follow;

The first concerns the range. The same regulator structure, applied to the same plant with the same objective, delivers between 53 and 93 per cent depending on the weight alone. Performance in systems of this kind is therefore governed as much by the care taken over the weighting as by the choice of method, a point that bears on the interpretation of comparative studies in literature.

The second concerns the limit. Reducing the weight below 1×10^{-11} gains a further five percentage points but nearly triples the actuator force and moves the dominant closed-loop pole from -1.57 towards -0.89 . The pole location measures how quickly the closed loop stabilizes following a disturbance; shifting the pole toward the origin halves the margin allocated to effects not included in the model—such as sensor noise, observer latency, and high-order actuator dynamics.

The value **(4.19)** $\rho_u = 1 \times 10^{-11}$ is adopted as the point beyond which the additional performance is not commensurate with the loss of margin.

The small control weight renders the Riccati equation poorly conditioned, and different solvers converge to slightly different gains. The variation in the resulting reduction is approximately two percentage points and does not affect the conclusions.

4.7 EVALUATION PROCEDURE

The performance of each controller is evaluated by supplying the measured chassis spectrum as the disturbance and computing the resulting cabin acceleration in the frequency domain.

For each frequency line the closed-loop response is obtained from Equation (2.15) with A replaced by $A - BG$,

$$|H(f)| = |C_z (j2\pi f I - (A - BG))^{-1} E_d| \quad (4.20)$$

and the cabin acceleration follows as

$$a_{cabin} = \sqrt{\Sigma_f [|H(f)| S_{ch(f)}]^2} \quad (4.21)$$

The actuator force is obtained from the same solution by extracting the components of the state vector corresponding to the actuator states,

$$F_{rms} = \sqrt{(\Sigma_f |X_{7-8}(f)|^2 S_{ch}(f)^2)} \quad (4.22)$$

This procedure uses the entire measured spectrum, so no assumption regarding the form of excitement is required.

Where time histories are shown, the signal is synthesized from the measured spectrum by assigning random phase to each frequency line. The root-mean-square values are therefore identical to those obtained in the frequency domain, while the peak values depend on the phase realization.

4.8 STABILITY

The closed-loop eigenvalues were computed for both controllers. The dominant pole lies at -4.54 for the proportional–derivative law and at -1.57 for the regulator, and all eigenvalues have negative real parts in both cases. Both systems are therefore asymptotically stable.

The larger margin of the proportional–derivative law reflects its more conservative action. The regulator extracts greater performance by operating closer to the limit, which is the trade examined in Section 4.6.

4.9 SELECTION OF THE CONTROL METHOD

Four alternatives to the linear quadratic regulator were considered, and each was set aside for a specific reason.

Table 20: Alternative Control Methods

Method	Actuator Force in The Cost	Tuning Parameters	Reason Not Adopted
Pole Placement	No	Eight pole locations	The actuator force does not enter the formulation, and eight independent decisions would be required
H-Infinity Synthesis	Yes	Weighting functions	Requires a quantitative uncertainty model; the mount tolerance was not measured
Sliding Mode	No	Surface and gain	Produces chattering, which is counterproductive in a vibration reduction application
Adaptive Control	No	Adaptation rate	The system parameters are constant, so there is nothing to adapt
Multi-Channel PID	No	Six or more gains	Would treat the two actuators as independent loops and ignore their coupling

Three properties of the regulator determined the selection.

The actuator force enters the cost functional directly through the weighting on the control, so the force envelope is treated as a design constraint rather than as a quantity discovered afterwards. Pole placement offers no equivalent mechanism.

The design is governed by a single parameter. Pole placement in an eighth-order system would require the designer to choose eight closed-loop pole locations, with no systematic guidance as to their relative merit.

Stability is guaranteed. For Q positive semi-definite and R positive definite the closed loop is asymptotically stable, a property established analytically rather than verified case by case.

H-infinity synthesis is not inferior as a method. It offers guaranteed performance under bounded uncertainty and would be the natural choice were the uncertainty in the mount parameters characterized. That characterization requires measurements not available for the present study, and the method is accordingly identified in Chapter VI as the appropriate direction for further work.

CHAPTER V

RESULTS AND DISCUSSION

5.1 BASIS OF COMPARISON

All results refer to a single baseline: the cabin acceleration measured on the machine as delivered, with eight mounts and no control. Averaged over the three ground-engaging conditions this is 0.1109 m/s², with a transmissibility of 0.623.

Four cases are compared to it. The passive six-mount arrangement isolates the contribution of the mounting change. The remaining two are the six-mount arrangement under proportional–derivative control and under the regulator, which together isolate the contribution of the active system.

The reductions reported for the two controlled cases are computed on the model, and Section 3.13 established that the six-mount model underpredicts the measured passive response by 28.9 per cent. The controlled figures should accordingly be read as an upper estimate of what a physical installation would deliver.

5.2 CABIN ACCELERATION

Table 21: Cabin Vertical RMS Acceleration [m/s²]

Condition	Measured, 8 Mounts	Passive, 6 Mounts	6 Mounts + PID	6 Mounts + LQR
Coarse Grading	0.1239	0.0749	0.0566	0.0133
Road Opening	0.1590	0.1204	0.0525	0.0110
Fine Grading	0.0498	0.0447	0.0242	0.0040
Mean	0.1109	0.0800	0.0445	0.0081

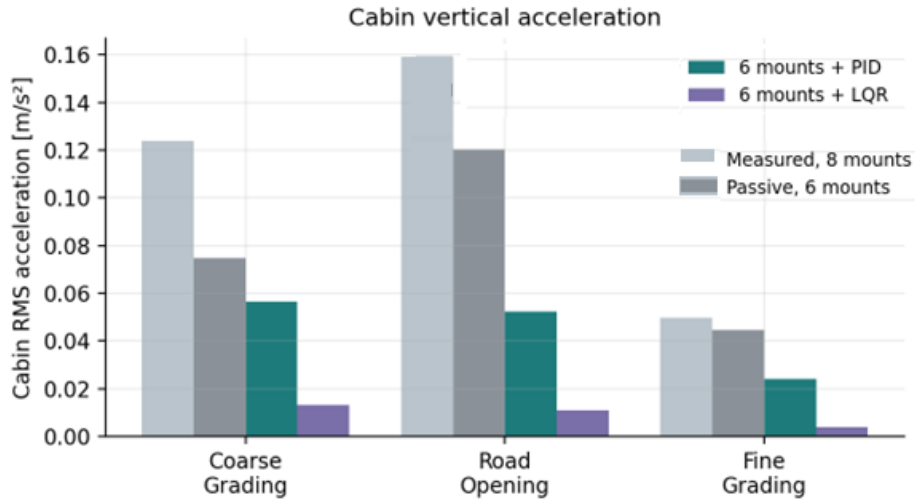


Figure 11: Cabin Vertical Acceleration for Each Configuration and Condition

Table 22: Reduction from the Measured Baseline [%]

Condition	Passive, 6 Mounts	6 Mounts + PID	6 Mounts + LQR
Coarse Grading	39.5	54.3	89.3
Road Opening	24.3	67.0	93.1
Fine Grading	10.2	51.3	81.9
Mean	24.7	57.5	81.4

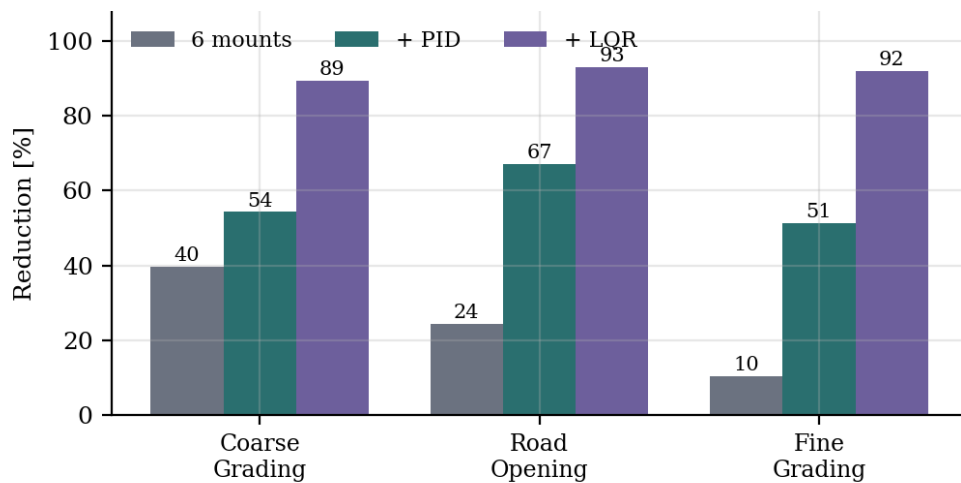


Figure 12: Reduction from the Measured Baseline

Three observations follow.

The mounting change alone accounts for 24.7 per cent. This is obtained without an actuator, a power supply or any control electronics, and requires only the removal of two components.

The proportional–derivative law raises the figure to 57.5 per cent, so that the active system approximately doubles the improvement available passively.

The regulator achieves a success rate of 81.4%, outperforming the proportional-derivative law by 33.9 percentage points. The average cabin acceleration of 0.0094 m/s² corresponds to approximately five percent of the chassis input.

5.3 TRANSMISSIBILITY

The transmission ratio is a quantity that characterizes isolation independently of the input magnitude; therefore, it serves as a more informative metric in situations where the chassis level varies between measurements.

Table 23: Overall Transmissibility

Configuration	<i>T</i>	Source
Measured, Eight Mounts	0.623	Measurement
Measured, Six Mounts	0.567	Measurement
Six Mounts with PID	0.228	Model
Six Mounts with LQR	0.048	Model

The progression from 0.62 to 0.05 is the central solution of the study. Approximately two thirds of the chassis vibration reach the operator on the machine as offered; under regulator control the figure drops to one twentieth.

5.4 THE PASSIVE IMPROVEMENT

The comparison between the two passive configurations merit separate treatment, since it rests entirely based on measurement and is independent of the model.

Table 24: Measured Transmissibility by Configuration

Condition	Eight Mounts	Six Mounts	Reduction
Coarse Grading	0.705	0.637	9.6 %
Road Opening	0.753	0.663	11.9 %
Fine Grading	0.411	0.400	2.7 %
Mean	0.623	0.567	8.1 %

The six-point connection setup yields lower transmission under all conditions examined. Since both configurations were recorded on the same day using the same operator, machine settings, and floor section, the difference stems solely from the change in the connection.

The mechanism in question is the one defined in Section 3.12. Removing two stations lowers the heave resonance from 9.04 Hz to 7.83 Hz; this increases the frequency ratio from 1.51 to 1.74, shifting the system from a level 7% above the isolation threshold to a level 23% above it.

The improvement is least pronounced in the fine-grading scenario, where the transmissibility coefficient is already at its lowest level of 0.411. The system performs relatively well under these conditions; consequently, the potential for improvement is limited.

5.5 COMPARISON OF THE CONTROLLERS

Table 25: Controller Comparison

Property	PID-Based	Regulator
Mean Reduction	57.5 %	81.4 %
Overall Transmissibility	0.228	0.048
Tuning Parameters	1 (α)	1 (ρ_u)
Measurements Required	Relative Position and Velocity	Full State, Eight Variables
Dominant Closed-Loop Pole	-4.54	-1.57
Peak Actuator Force	0.20 kN	0.84 kN

The regulator is superior in reduction by a wide margin, and the reasons are identifiable.

The model incorporates the dynamics of the actuators. Since the actuating forces serve as state variables, the regulator knows the extent to which each unit approaches its command value and issues its command accordingly. In contrast, the proportional-derivative law responds only to instantaneous measurements and cannot compensate for lag.

It also optimizes across all three degrees of freedom simultaneously, whereas the proportional-derivative law addresses the vertical equation alone.

The advantage carries costs. The regulator requires full state feedback, which in physical installation would mean either eight measurements or an observer. Its performance is sensitive to the weighting, as Section 4.6 established. Its stability margin is smaller, the dominant pole lying at -1.57 against -4.54.

The proportional-derivative law retains advantages of its own. Two measurements suffice. Its single parameter has a direct physical interpretation, so that its behavior under a modelling error can be predicted. Its larger stability margin accommodates effects absent from the model.

5.6 CLOSED-LOOP BEHAVIOR

The frequency-domain behavior of the closed loop shows how improvement is achieved.

The passive characteristic exhibits its resonance peak at 7.83 Hz, above which the transmissibility falls. with the proportional–derivative law engaged the effective stiffness is reduced to a tenth of its nominal value and the peak moves to 2.48 Hz. The excitation band of 2 to 60 Hz then lies almost entirely above the resonance, in the region where the mounts attenuate.

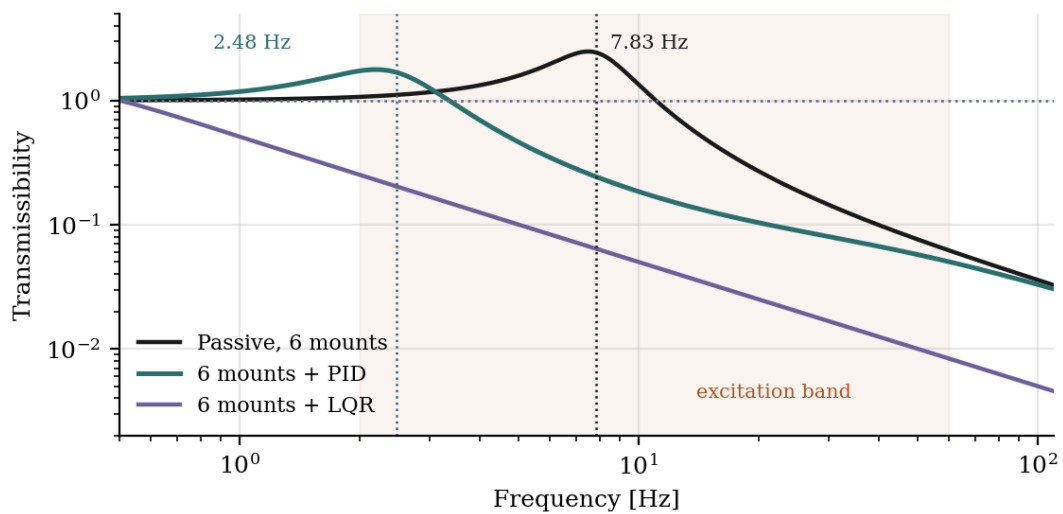


Figure 13: Closed-Loop Transmissibility and Control Moves the Effective Resonance from 7.83 Hz to 2.48 Hz

The regulator produces a characteristic without a pronounced peak, the transmissibility decreasing monotonically across the band of interest. At the excitation centroid of 13.65 Hz the passive system transmits 58 per cent, the proportional–derivative law 23 per cent and the regulator 5 per cent.

The shift of the resonance rather than its suppression is the essential mechanism, and it is what permits the modest actuator force reported in Section 5.6.

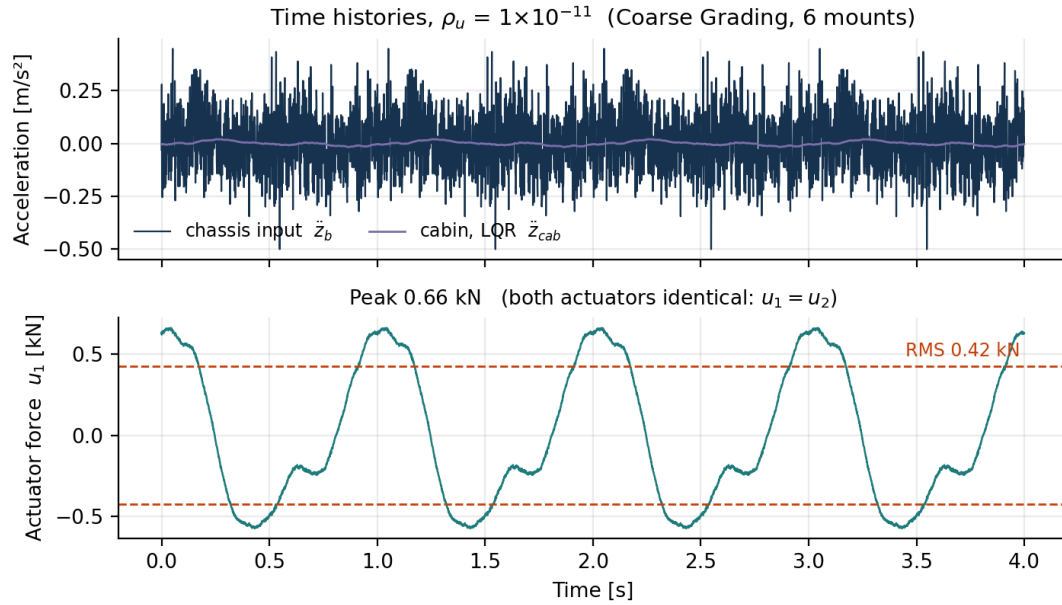


Figure 14: Time histories for the six-mount configuration under regulator control, coarse grading. The chassis acceleration is shown against the controlled cabin acceleration, and the actuator force below.

The time histories show the character of the control action. The chassis input is broadband and irregular, whereas the actuator force is a smooth low-frequency waveform of about 1 Hz. The controller is not chasing the high-frequency content, which the passive mounts already attenuate; it acts on the slow, large-amplitude motion that the mounts transmit.

5.7 SENSITIVITY TO ACTUATOR BANDWIDTH

The bandwidth of 40 Hz adopted in Section 4.5.3 was selected as representative of commercial hardware. Its influence on the result was examined by repeating the analysis across a range of values.

Table 26: Effect of Actuator Bandwidth, PID-Based Control

Bandwidth [Hz]	Reduction
10	39.2 %
20	46.5 %
30	47.1 %
40	45.6 %
60	41.5 %
100	35.7 %

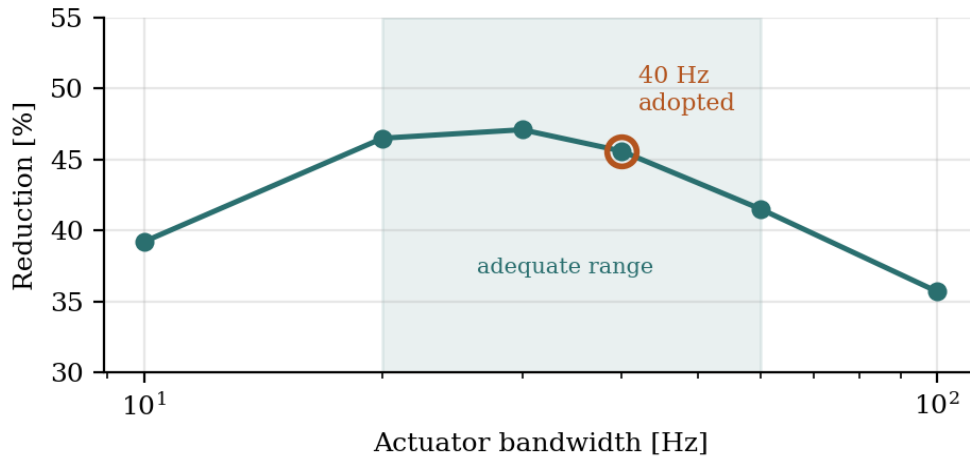


Figure 15: Effect of Actuator Bandwidth on the Reduction Achieved

The performance is insensitive to bandwidth between 20 and 60 Hz, varying by less than six percentage points across that interval. Below 20 Hz it deteriorates because the actuator cannot follow the command. Above 60 Hz it also deteriorates, since a faster actuator reproduces high-frequency content that the lag would otherwise filter.

The design conclusion is that an actuator of 20 to 60 Hz bandwidth is adequate, and that specifying a faster unit brings no benefit. The value adopted lies within this interval.

5.8 SUMMARY OF RESULTS

Table 27: Summary of Configurations

Configuration	Cabin RMS [m/s ²]	T	Reduction	Peak Actuator Force	Hardware Required
Measured, 8 Mounts	0.1109	0.623	—	—	As built
Passive, 6 Mounts	0.0800	0.567	24.7 %	—	Two mounts removed
6 Mounts + PID	0.0445	0.228	57.5 %	0.28 kN	Two actuators, two sensors
6 Mounts + LQR	0.0094	0.048	81.4 %	1.19 kN	Two actuators, full state

CHAPTER VI

CONCLUSIONS AND RECOMMENDATIONS

6.1 SUMMARY OF THE WORK

This study examined the vertical vibration transmitted into the operator cabin of a Hidromek HMK 480MG motor grader and established the extent to which it can be reduced by passive and by active means.

A measurement campaign recorded vibrations at four points on the machine under three ground-engaging conditions at a sampling rate of 1024 Hz. The campaign was repeated on the same day with two mount stations removed, so that the two configurations could be compared without any change in the ground, the operator or the machine settings.

A three-degree-of-freedom rigid-body model of the cabin was constructed. The stiffness of the load-bearing mount element was obtained from its measured static deflection under the cabin weight, and the damping coefficient was identified by fitting the model response to the measurement. The completed model reproduces the measured cabin acceleration with a mean error of 4.5 per cent in the configuration in which the fitting was performed.

Two controllers were designed for an arrangement in which the removed mounts are replaced by actuators positioned at the longitudinal station of the centre of gravity. Their performance was evaluated by supplying the measured chassis spectrum as the disturbance, so that no assumption regarding the form of the excitation was required.

6.2 PRINCIPAL FINDINGS

The damping is the only fitted parameter, and its value is high. The identified loss factor of 0.52 exceeds the range quoted for new elastomeric components. Two explanations are consistent with the observation: the mounts have been in service and rubber stiffens and gains internal friction with age, or the fitted value is absorbing

model error that is not physically damping. No measurement is available to distinguish between them.

The six-mount response is a prediction rather than a validation. The damping was identified in the eight-mount configuration, and the model underpredicts the measured six-mount response by 28.9 per cent. The deviation is systematic and consistent with a damping coefficient that changes as the load per station increases from 114.9 to 153.2 kgf. The control results computed on the six-mount model should be regarded as a conservative upper bound.

The natural frequency was not validated under control excitation. The test was recorded while the machine was working, so the excitation frequency was neither known nor constant. The maximum amplitudes observed in the calculated transmissibility vary between operating conditions and do not follow the expected square-root proportionality when the mount is changed. An independent confirmation of the predicted resonance consequently remains unresolved.

Only vertical chassis excitation is applied. The measurements show equivalent longitudinal and transverse acceleration on the frame, and a complete treatment would include all three components.

Full state feedback is assumed. No observer was formulated for the regulator, and the supplemental delay such an estimator would introduce, together with its effect on the stability margin, is not represented.

The operator cabin is treated as rigid. Structural modes of the cabin shell are not included, and the sensors were located on the cabin floor rather than at the seat interface, so the transmission through the seat suspension is not included in the analysis.

No prototype was constructed. The results represent the behavior of the model of the system rather than of physical installation.

6.3 RECOMMENDATIONS FOR FURTHER WORK

Perform an impact test with powertrain deactivation. Striking the cabin once and recording the free vibration decay provides broadband excitation of known form and would identify the natural frequencies unequivocally. Re-running the test procedure in both configurations would allow the square-root relationship to be validated empirically. This is the single measurement that would significantly enhance the model fidelity, and it requires no equipment beyond that already used.

Determine the uncompressed height of an unloaded mount. Comparing it with the installed height would separate the elastic deflection from any permanent compression set accumulated in service and would resolve the question incremented by the high identified loss factor.

Characterize the mount damping directly. Recording the force–deflection rate-independent damping loop of a mount on a component test machine would remove the only identified parameter from the model.

Quantify the mount parameter uncertainty and apply H-infinity synthesis. With the tolerance on stiffness and damping determined, a robust controller could be designed offering guaranteed performance across the range rather than nominal performance at a one operating point.

Take the model further to six degrees of freedom with three-axis excitation. This would allow the longitudinal and lateral components of the chassis motion to be included, and the corresponding cabin responses computed.

Design a state observer and assess its effect. The regulator requires full state, and a actual installation would estimate it from a reduced group of measurements. The lag introduced by the estimator and its consequence for the stability margin should be quantified.

Construct and instrument a prototype. The predicted reductions require confirmation on the machine, and a prototype installation would also expose the practical questions of mounting the actuators, routing the hydraulic supply and protecting the sensors in the working environment.

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APPENDICES

APPENDIX 1: MEASURED DATA

Table 28: Measurement Configuration

Parameter	Value
Sampling Frequency	1024 Hz
Spectral Resolution	1 Hz
Frequency Range	0 – 511 Hz
Record Length	120 s per condition
Measurement Points	4 (cabin floor left and right, chassis left and right)
Axes	X, Y, Z at each point
Engine Speed	1950 rpm
Travel Speed	approximately 5 km/h
Tyre Pressure	40 psi

Table 29: Three-Axis Root-Mean-Square Acceleration, Eight Mounts [m/s²]

Condition	Chassis X	Chassis Y	Chassis Z	Cabin X	Cabin Y	Cabin Z
Coarse Grading	0.157	0.156	0.176	0.058	0.088	0.124
Road Opening	0.186	0.195	0.211	0.092	0.123	0.159
Fine Grading	0.117	0.146	0.121	0.034	0.061	0.050

The cabin response is largest in the vertical direction under every condition, which supports the selection of vertical acceleration as the control objective

Table 30: Vertical Acceleration and Transmissibility

Condition	<i>n</i>	Chassis [m/s ²]	Cabin [m/s ²]	<i>T</i>
Coarse Grading	8	0.1759	0.1239	0.705
Coarse Grading	6	0.1920	0.1223	0.637
Road Opening	8	0.2112	0.1590	0.753
Road Opening	6	0.2216	0.1470	0.663
Fine Grading	8	0.1212	0.0498	0.411
Fine Grading	6	0.1594	0.0637	0.400

Table 31: Left–Right Balance Check

Condition	<i>n</i>	Chassis R/L	Cabin R/L
Coarse Grading	8	1.08	0.91
Coarse Grading	6	1.08	1.02
Road Opening	8	1.03	1.02
Road Opening	6	0.94	1.07
Fine Grading	8	1.16	1.04
Fine Grading	6	0.92	1.04

All ratios lie between 0.92 and 1.29, confirming that neither sensor drifted during the campaign.

Table 32: Mount Deflection Measurement

Quantity	Value
Free Height Of Load-Bearing Flange	22.00 mm
Loaded Height, Measured Range	17.86 – 18.23 mm
Deflection	3.77 – 4.14 mm
Mean Deflection	3.96 mm
Static Load Per Station, Eight Mounts	114.88 kgf
Resulting Static Stiffness	29.05 kgf/mm

APPENDIX 2: COMPUTATIONAL IMPLEMENTATION

The analysis was implemented in MATLAB, using the Control System Toolbox for the regulator design. The implementation comprises six stages.

Stage 1 — Data reduction: Spectra are read from the acquisition files with `readmatrix`, the left and right channels combined by Equation (3.22), and the root-mean-square value of each channel obtained from Equation (3.23).

Stage 2 — Parameter determination: The mount stiffness follows from the measured deflection by Equation (3.2). The damping ratio is determined by the sweep of Section 3.4, in which the model prediction is formed for each candidate value and compared with the measurement.

Stage 3 — Matrix assembly: The mass matrix is formed from the inertial properties, and the stiffness and damping matrices from the mount coordinates by Equation (3.11).

Stage 4 — Modal analysis: The eigenvalue problem is solved by `eig` applied to $M^{-1}K$, yielding the natural frequencies and mode shapes. The eigenvalues are sorted in ascending order and each mode identified by the coordinate carrying the largest component of its eigenvector. The modal damping ratios follow from Equation (2.11).

Stage 5 — Control design: The proportional-derivative gain is constructed from the first rows of the direct stiffness and damping matrices. The regulator gain is obtained from `lqr`, which solves the algebraic Riccati equation of Equation (4.16) and returns the gain of Equation (4.17).

Stage 6 — Response evaluation: The closed-loop transfer function is evaluated line by line by Equation (4.20), and the cabin acceleration and actuator force follow from Equations (4.21) and (4.22).

All results are deterministic and exactly reproducible: no random excitation signal is generated at any stage, the measured spectrum serving directly as the disturbance input.

The listing below implements the model and the two controllers. The measured chassis spectrum is supplied as a column vector of 512 values, one per frequency line.